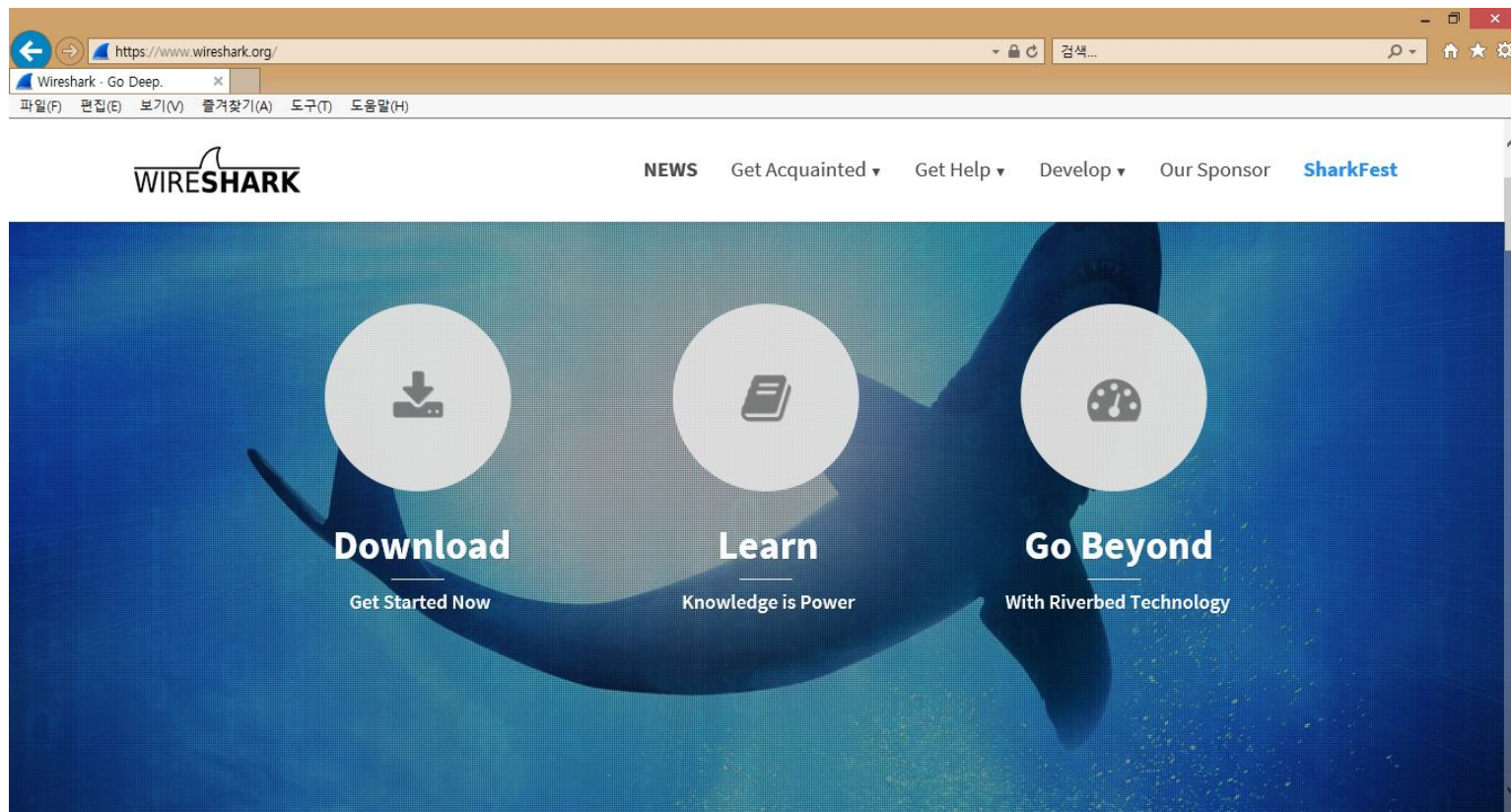


1. Sniffing 환경구성

Sniffing 환경구성

- 패킷 스니핑(sniffing) 또는 프로토콜 분석(protocol analysis) 툴
- 네트워크를 통해 전달되는 데이터를 수집하고 해석



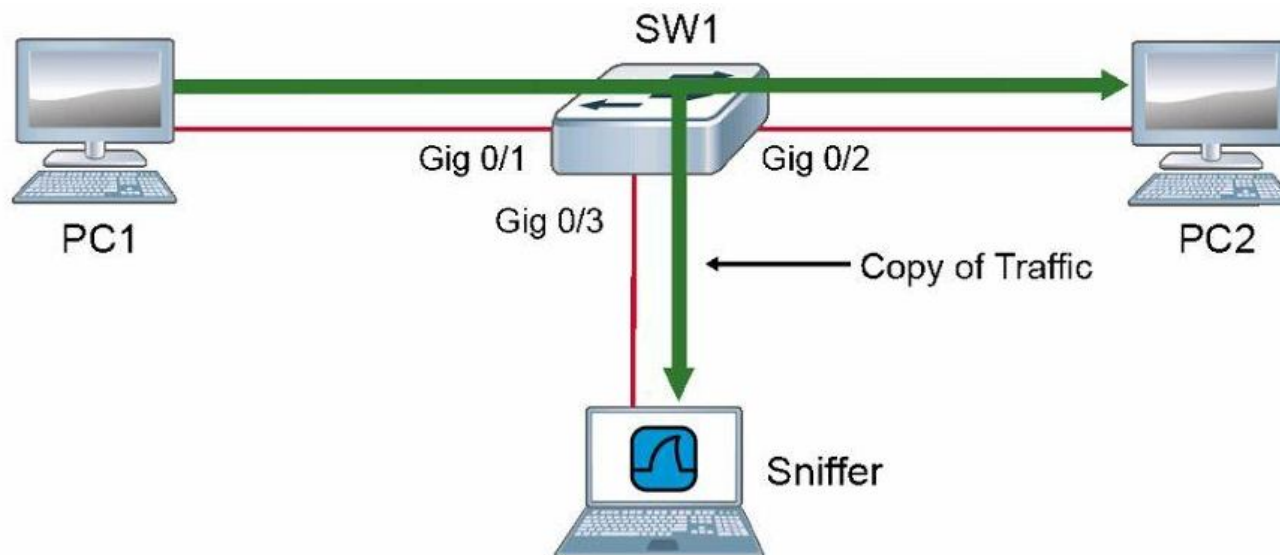
<http://www.wireshark.org>

* 패킷 수집과 패킷 분석

	LINUX	MS-Window
패킷수집	TCPdump	TShark
패킷분석	Wireshark	

1) 무차별모드(Promiscuous mode)

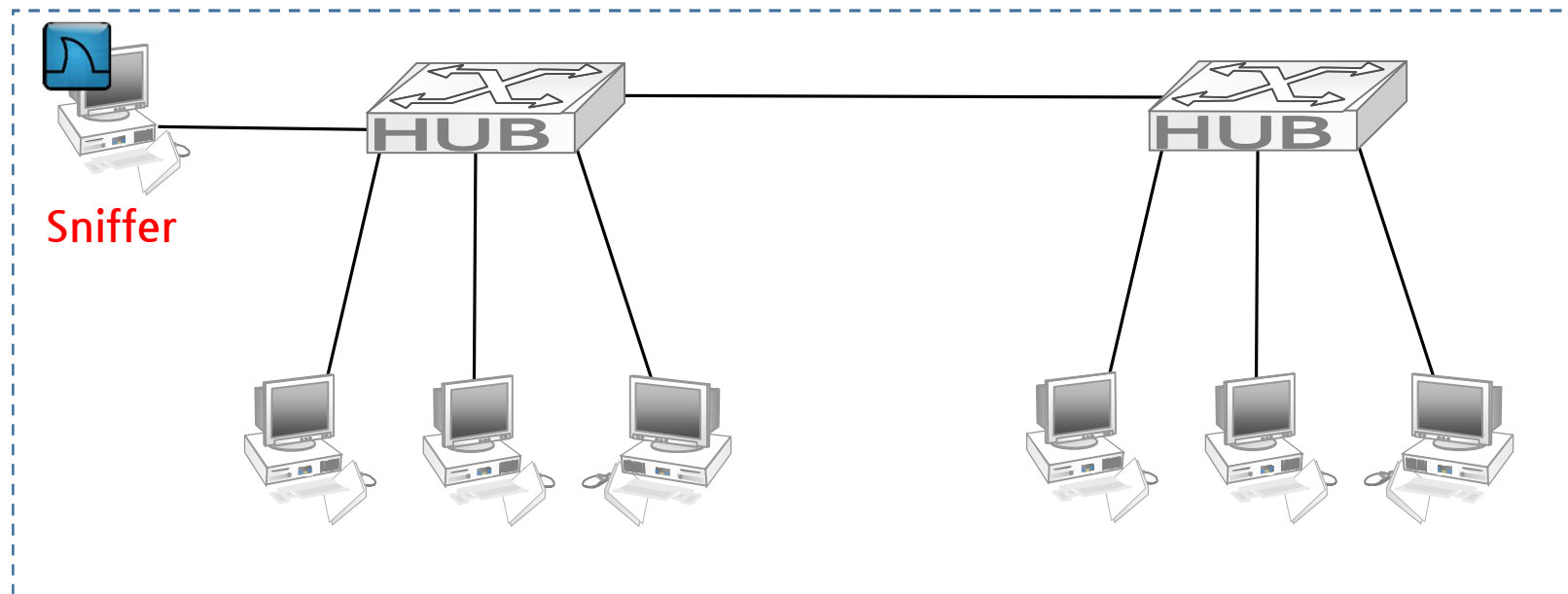
- 데이터 링크 계층과 네트워크 계층의 주소 필터링을 해제한 모드
- 주소 지정에 관계없이 호스트의 프로세서에 모든 패킷을 전달
- 패킷이 CPU에 전달되면 분석을 위해 패킷 스니핑 애플리케이션에 넘겨짐
- 유닉스 또는 리눅스에서는 'ifconfig eth0 promisc'로 설정
- Window에서는 npcap 설치로 무차별 모드 설정



2) Packet Sniffing 환경

① 허브 환경에서의 스니핑

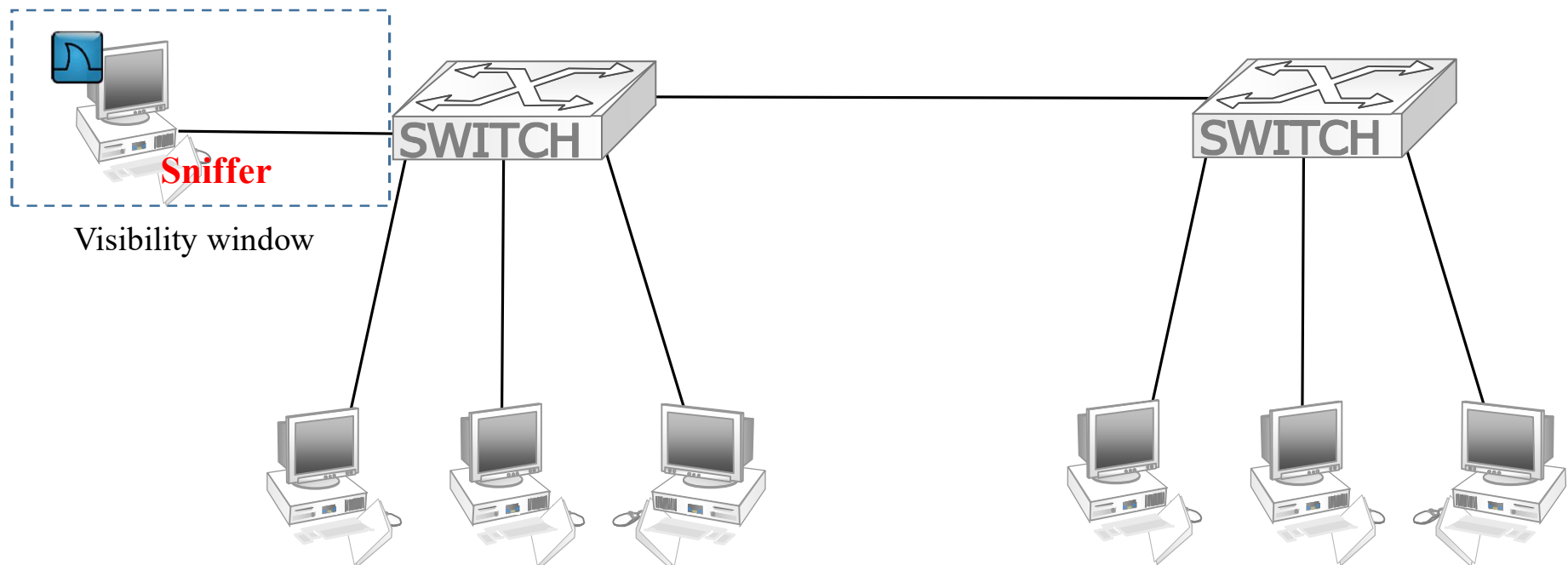
- 허브를 통해 전송되는 트래픽은 해당 허브에 연결된 모든 포트를 통과



Visibility window (가시성 창)

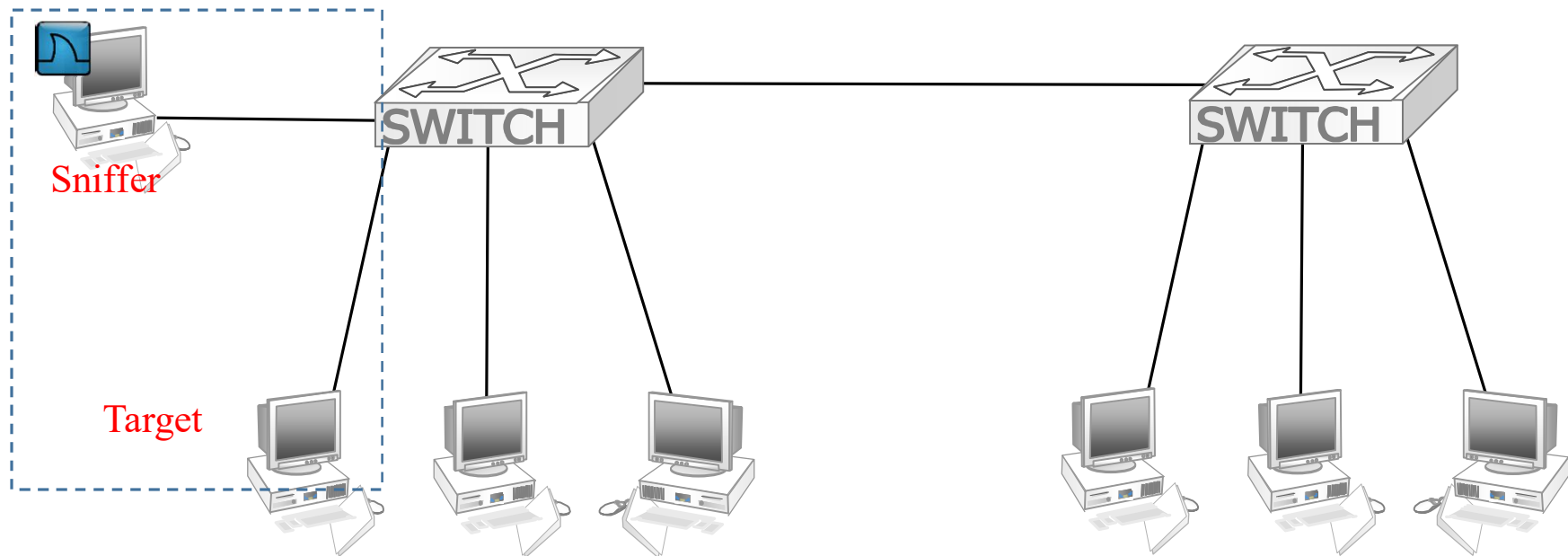
② 스위치 환경에서 스니핑 방법

- 포트 미러링(port mirroring)
- 허빙 아웃(hubbing out)
- 탭 사용 (tapping)



포트 미러링

Visibility window

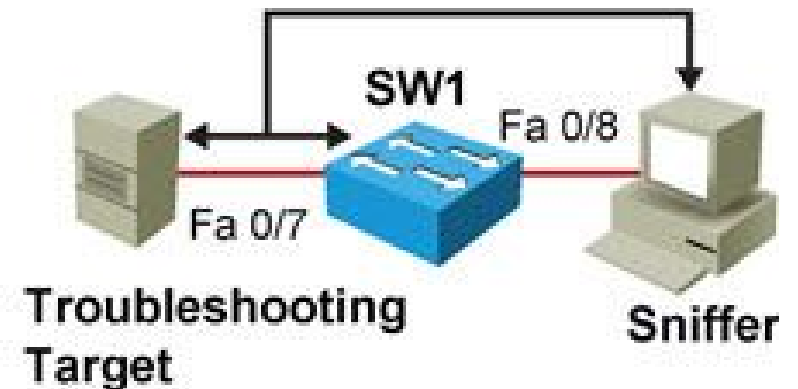


•Switched Port Analyzer (SPAN)

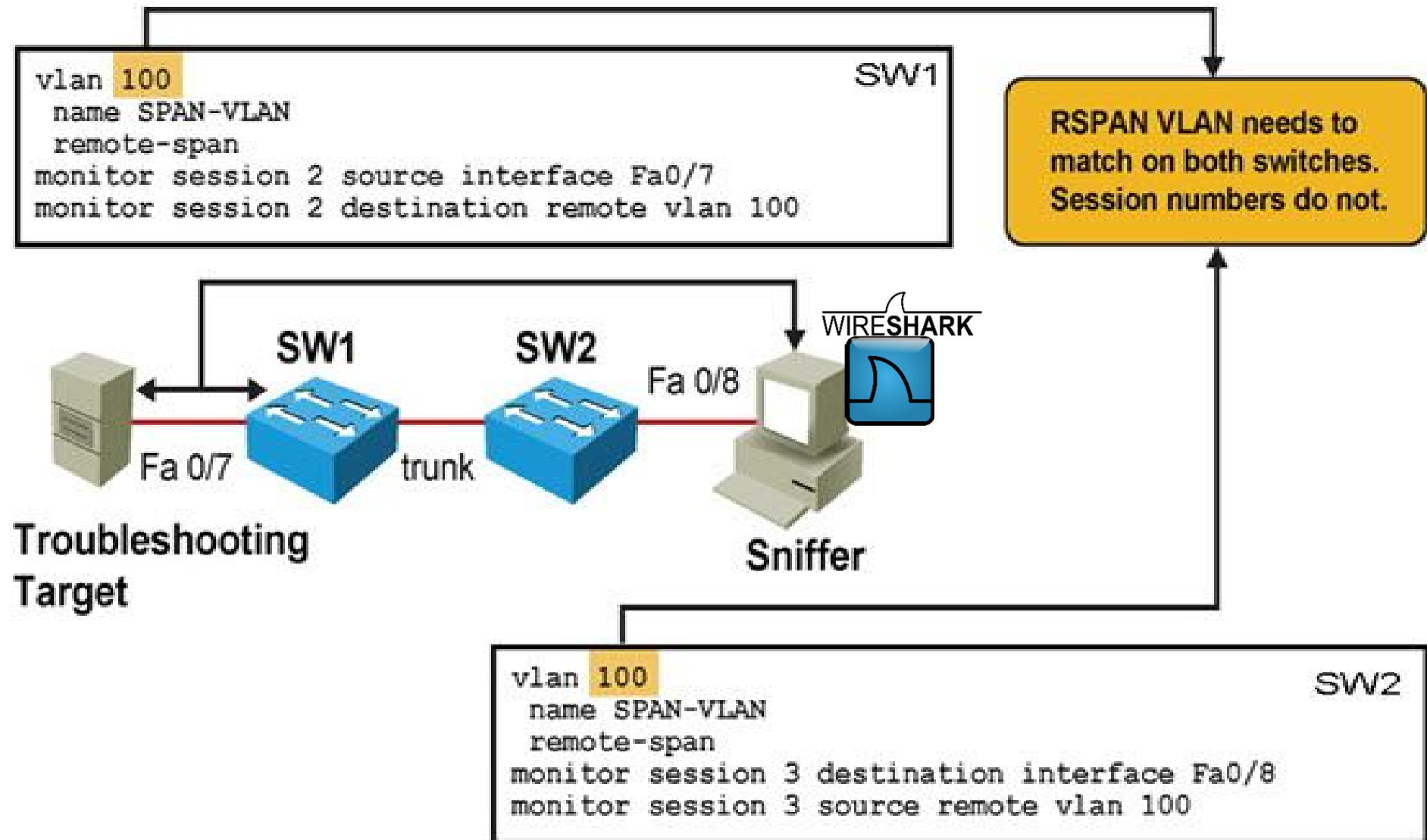
```
monitor session 1 source interface Fa0/7  
monitor session 1 destination interface Fa0/8
```

Sources and destinations that form a single SPAN session are identified by a session number

```
SW1#show monitor  
Session 1  
-----  
Type : Local Session  
Source Ports :  
    Both : Fa0/7  
Destination Ports : Fa0/8  
Encapsulation : Native  
Ingress : Disabled
```

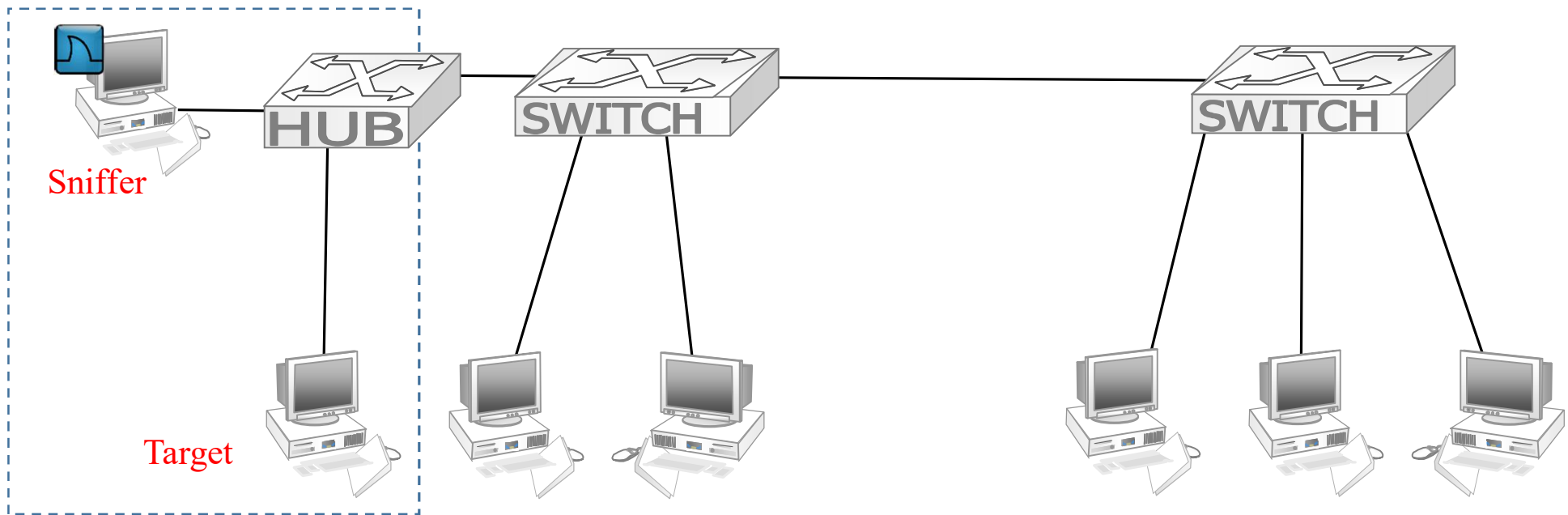


•Remote Switched Port Analyzer (RSPAN)



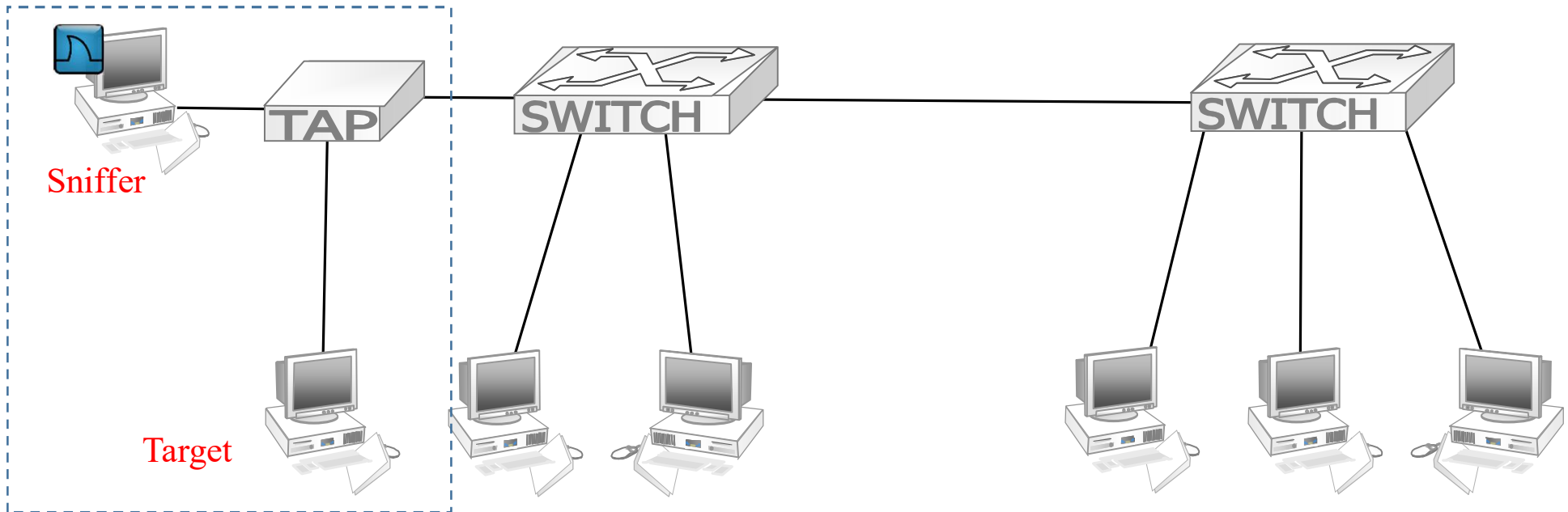
허빙 아웃(hubbing out)

Visibility window



탭 사용 (tapping)

Visibility window

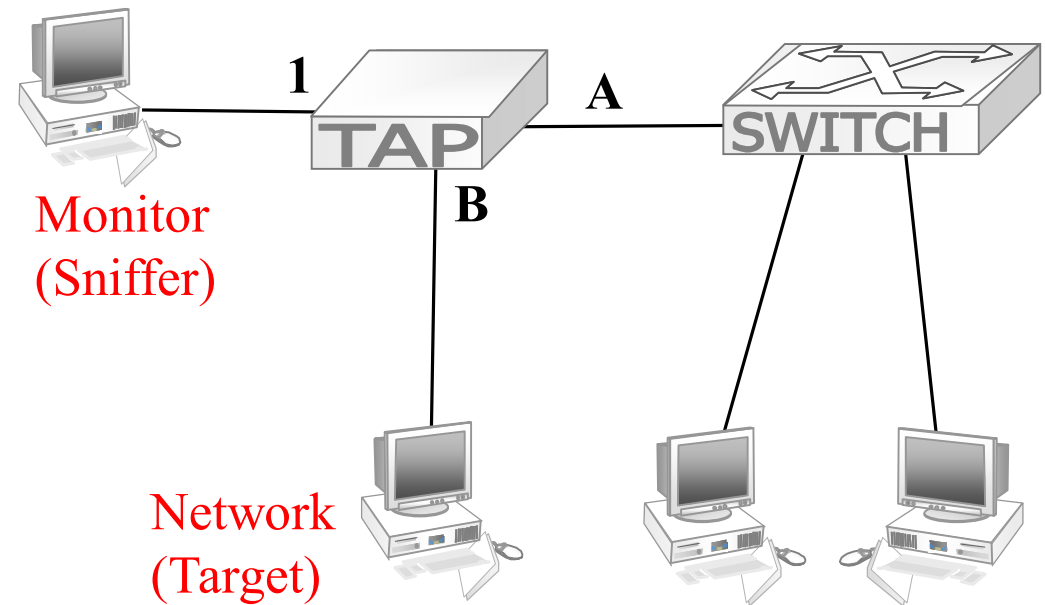
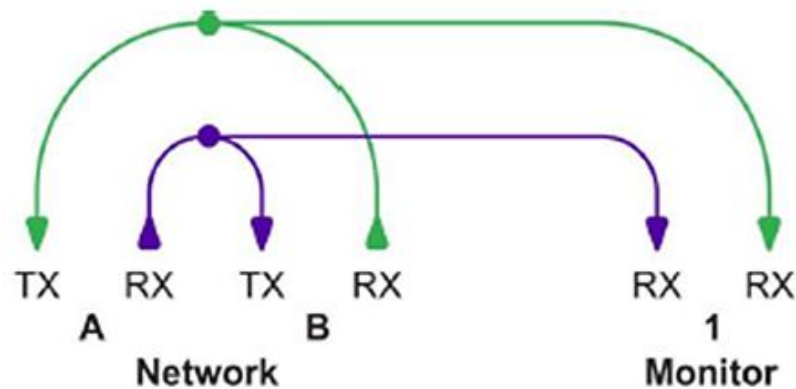


TAP(Test Access Ports)

- 데이터 흐름의 중단 없이, 네트워크에 전혀 영향을 주지 않으면서 트래픽을 모니터링 할 수 있게 해주는 모니터링 장비
- 최근 네트워크 탭은 스위치처럼 12포트, 24포트가 있어 많은 수의 장비를 동시에 사용가능

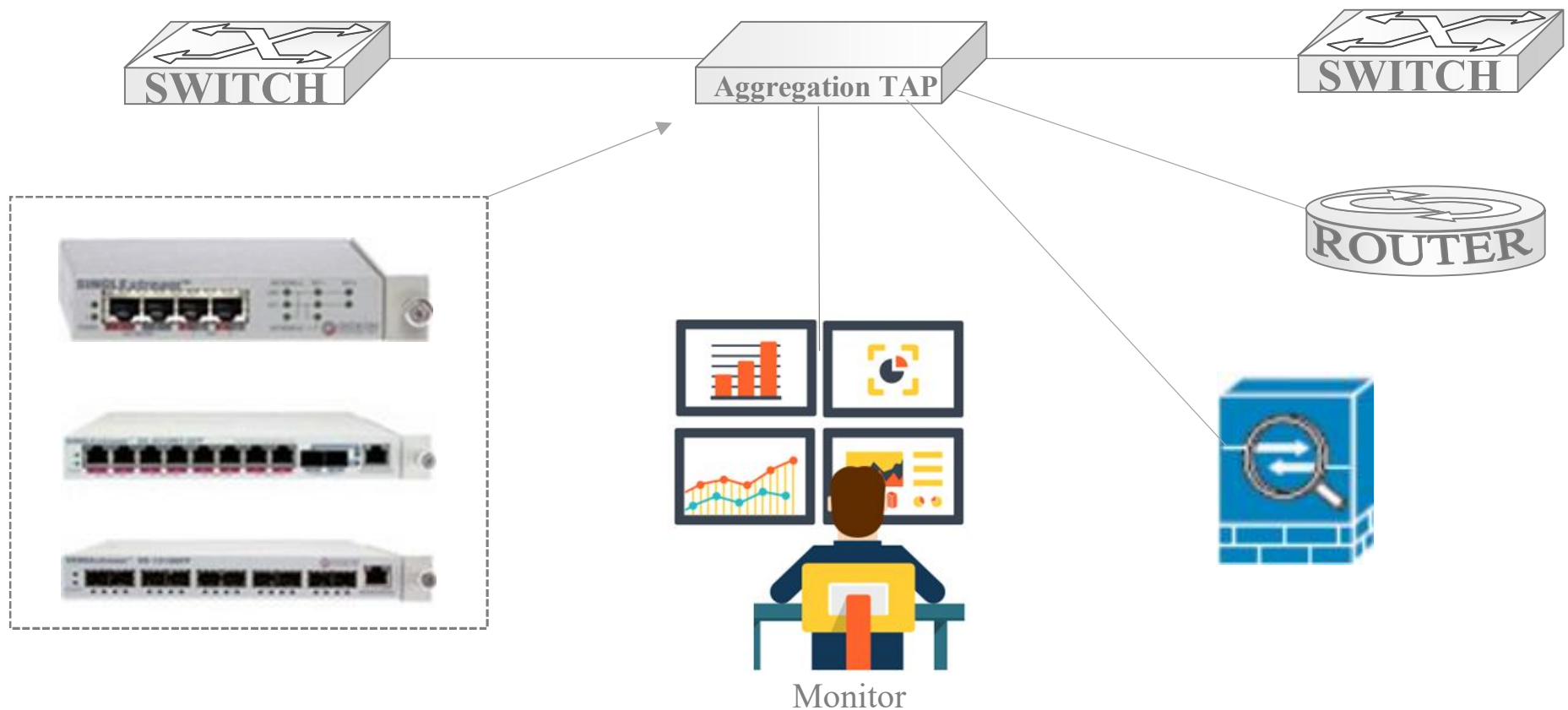
스위치 환경에서의 스니핑 - Aggregated Tap

- 3개의 포트 구성
- 양방향 트래픽을 스니핑하기 위한 물리적 모니터 포트를 하나만 갖고 있음



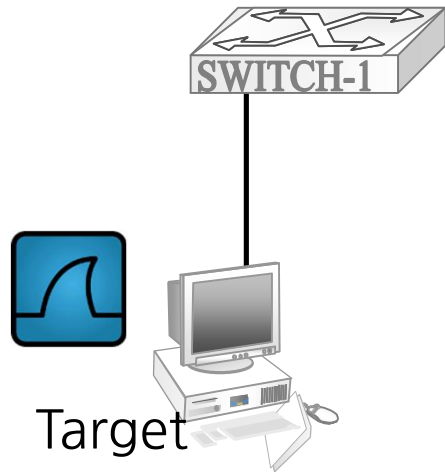
Aggregation TAP

- 1개 이상의 Network Links 또는 Ports에서 수집한 Packet Data들을, 1개의 NIC를 사용하는 Monitor 또는 그 이상의 여러 Monitor 장비로 보내 분석할 수 있게 해 주는 TAP

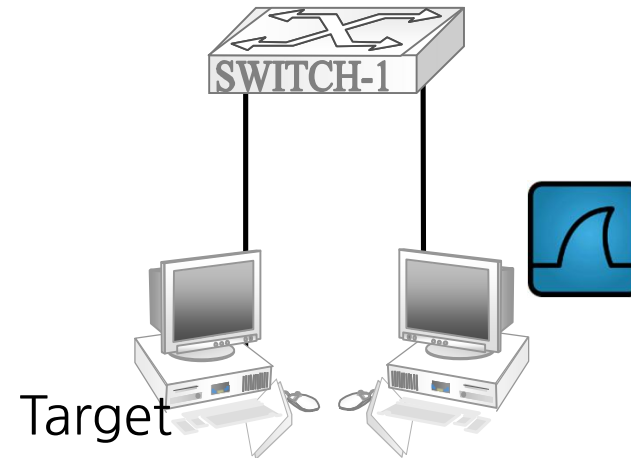


트래픽 수집 위치

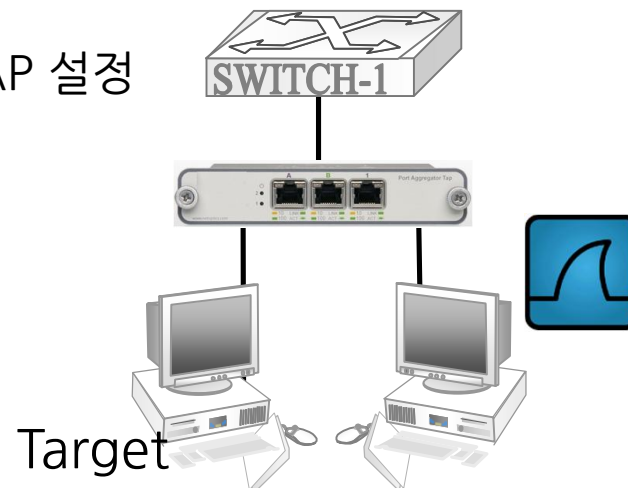
① 지연이 발생한 호스트에서 직접 수집



② 호스트의 스위치 포트를 확장(포트미러링)

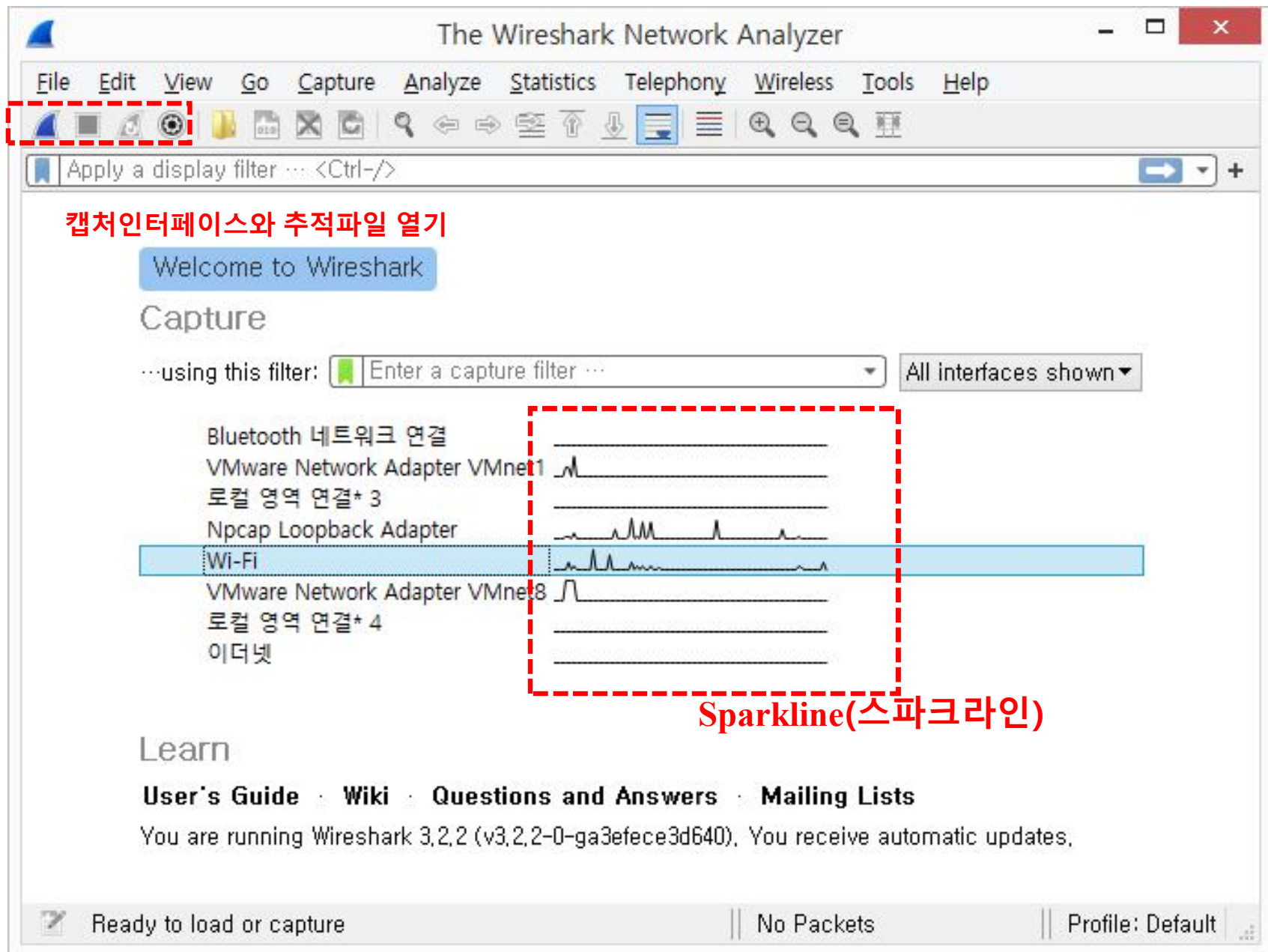


③ TAP 설정



2. Wireshark Tool 기능

2.1 Wireshark Tool



1 타이틀바(제목표시줄)

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
 > Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)
 > Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15
 ▾ Transmission Control Protocol
 Source Port: 51006 (51006)
 Destination Port: https (443)
 [Stream index: 0]
 [TCP Segment Len: 0]

0000	b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00	..O...h..Q.S..E.
0010	00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7	.4*..@...ML...E.
0020	90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02	...>.....
0030	fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01	
0040	04 02	..

Wi-Fi: <live capture in progress>
 Packets: 240 · Displayed: 240 (100,0%)
Profile: Default

Capturing from Wi-Fi

2 메인 메뉴

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Expression...

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x00
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x00
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR _

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

> Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)

> Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15

Transmission Control Protocol

Source Port: 51006 (51006)
Destination Port: https (443)
[Stream index: 0]
[TCP Segment Len: 0]

0000	b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00	..O...h..Q.S..E..
0010	00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7	.4*..@...ML...E..
0020	90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02	...>.....
0030	fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01	
0040	04 02	..

Wi-Fi: <live capture in progress>
Packets: 240 · Displayed: 240 (100.0%)
Profile: Default

본 교재는 수강생 개인 학습용으로만 제공됩니다. 무단 복제, 배포, 수정, 공유, 온라인 게시 행위를 금합니다.

18

Capturing from Wi-Fi

3

메인 툴바

File

Edit

View

Go

Capture

Analyze

Statistics

Telephony

Wireless

Tools

Help

Apply a display filter ... <Ctrl-/>

Expression...

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR _

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

> Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)

> Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15

▼ Transmission Control Protocol

Source Port: 51006 (51006)

Destination Port: https (443)

[Stream index: 0]

[TCP Segment Len: 0]

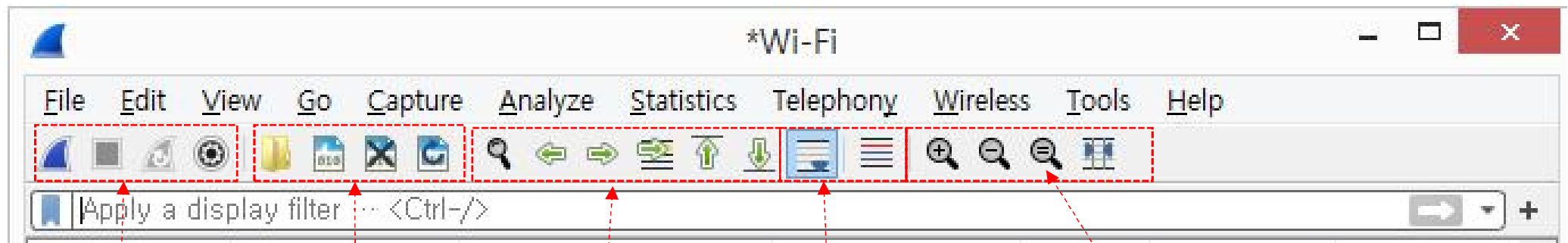
0000	b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00	..O...h..Q.S..E.
0010	00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7	.4*..@... ML...E.
0020	90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02	...>.....
0030	fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01	
0040	04 02	..

Wi-Fi: <live capture in progress>

Packets: 240 · Displayed: 240 (100.0%)

Profile: Default

본 교재는 수강생 개인 학습용으로만 제공됩니다. 무단 복제, 배포, 수정, 공유, 온라인 게시 행위를 금합니다.



컬러화

뒤/앞/지정된번호/처음/끝이동

Zoom-in
/out/100%
/Resize Columes

파일을 열기/저장/닫기/재로드

캡처을 시작/중지/재시작/캡션옵션

4 디스플레이 필터 영역과 필터 표현식 영역

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR _

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

> Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)

> Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15

▼ Transmission Control Protocol

Source Port: 51006 (51006)

Destination Port: https (443)

[Stream index: 0]

[TCP Segment Len: 0]

0000	b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00	..O...h..Q.S..E.
0010	00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7	.4*.@... ML...E.
0020	90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02	...>.....
0030	fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01	
0040	04 02	..

Wi-Fi: <live capture in progress> || Packets: 240 · Displayed: 240 (100.0%) || Profile: Default

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

5 패킷 목록(list) 창

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR _

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

> Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)

> Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15

▼ Transmission Control Protocol

Source Port: 51006 (51006)

Destination Port: https (443)

[Stream index: 0]

[TCP Segment Len: 0]

```

0000  b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00  ..O...h..Q.S..E.
0010  00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7  .4*.@...ML...E.
0020  90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02  ...>.....
0030  fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01  .....
0040  04 02  ..

```

Wi-Fi: <live capture in progress> | Packets: 240 · Displayed: 240 (100.0%) | Profile: Default

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

> Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)

> Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15

▼ Transmission Control Protocol

Source Port: 51006 (51006)

Destination Port: https (443)

[Stream index: 0]

[TCP Segment Len: 0]

0000 b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00 ..O...h..Q.S..E.

0010 00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7 .4*..@...ML...E.

0020 90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02 ...>.....

0030 fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01E.....

0040 04 02 ..

Wi-Fi: <live capture in progress> | Packets: 240 · Displayed: 240 (100.0%) | Profile: Default

⑥ 패킷 상세(details) 창

Capturing from Wi-Fi

File
Edit
View
Go
Capture
Analyze
Statistics
Telephony
Wireless
Tools
Help

Apply a display filter ... <Ctrl-/>
Expression...

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x0
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR _

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
> Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)
> Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15
▼ Transmission Control Protocol
Source Port: 51006 (51006)
Destination Port: https (443)
[Stream index: 0]
[TCP Segment Len: 0]

0000	b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00	..O...h...Q.S.E.
0010	00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7	.4*.@...ML...E.
0020	90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02	...>.....
0030	fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01	
0040	04 02	..

Wi-Fi: <live capture in progress>
Packets: 240 · Displayed: 240 (100.0%)
Profile: Default

⑦ 패킷 바이트(byte) 창

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Destination	Protocol	Length	Info
233	0.000000	172.30.1.29	172.30.1.255	DB-LSP-DISC	211	Dropbox LAN sync Discovery
234	3.044096	40.100.49.18	172.30.1.15	TLSv1.2	97	Application Data
235	0.000001	40.100.48.98	172.30.1.15	TLSv1.2	97	Application Data
236	0.051135	172.30.1.15	40.100.48.98	TCP	54	50756 → https(443) [ACK] Seq
237	0.000001	172.30.1.15	40.100.49.18	TCP	54	50747 → https(443) [ACK] Seq
238	0.693244	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x00
239	1.023859	172.30.1.12	224.0.0.251	MDNS	222	Standard query response 0x00
240	0.001810	fe80::1021:ec19:c0...	ff02::fb	MDNS	292	Standard query 0x0000 PTR _

< >

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

> Ethernet II, Src: IntelCor_51:0b:53 (68:ec:c5:51:0b:53), Dst: Mercury_15:eb:de (b4:a9:4f:15:eb:de)

> Internet Protocol Version 4, Src: 172.30.1.15, Dst: 69.167.144.15

▼ Transmission Control Protocol

Source Port: 51006 (51006)

Destination Port: https (443)

[Stream index: 0]

[TCP Segment Len: 0]

0000 b4 a9 4f 15 eb de 68 ec c5 51 0b 53 08 00 45 00 ..O...h..Q.S..E.

0010 00 34 2a 94 40 00 80 06 4d 4c ac 1e 01 0f 45 a7 .4*..@...ML...E.

0020 90 0f c7 3e 01 bb d9 96 8b f3 00 00 00 00 80 02 ...>.....

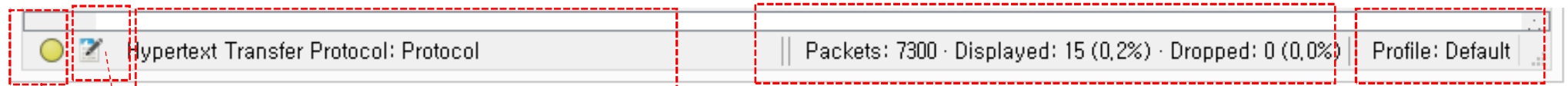
0030 fe 88 bf 1f 00 00 02 04 05 b4 01 03 03 08 01 01

0040 04 02 ..

8 상태바(Status Bar)

Wi-Fi: <live capture in progress> | Packets: 240 · Displayed: 240 (100.0%) | Profile: Default

패킷 개수와 로드 시간 정보



필드, 수집 또는 추적 파일 정보

인용버튼

현재 사용중인 프로파일

전문가정보버튼

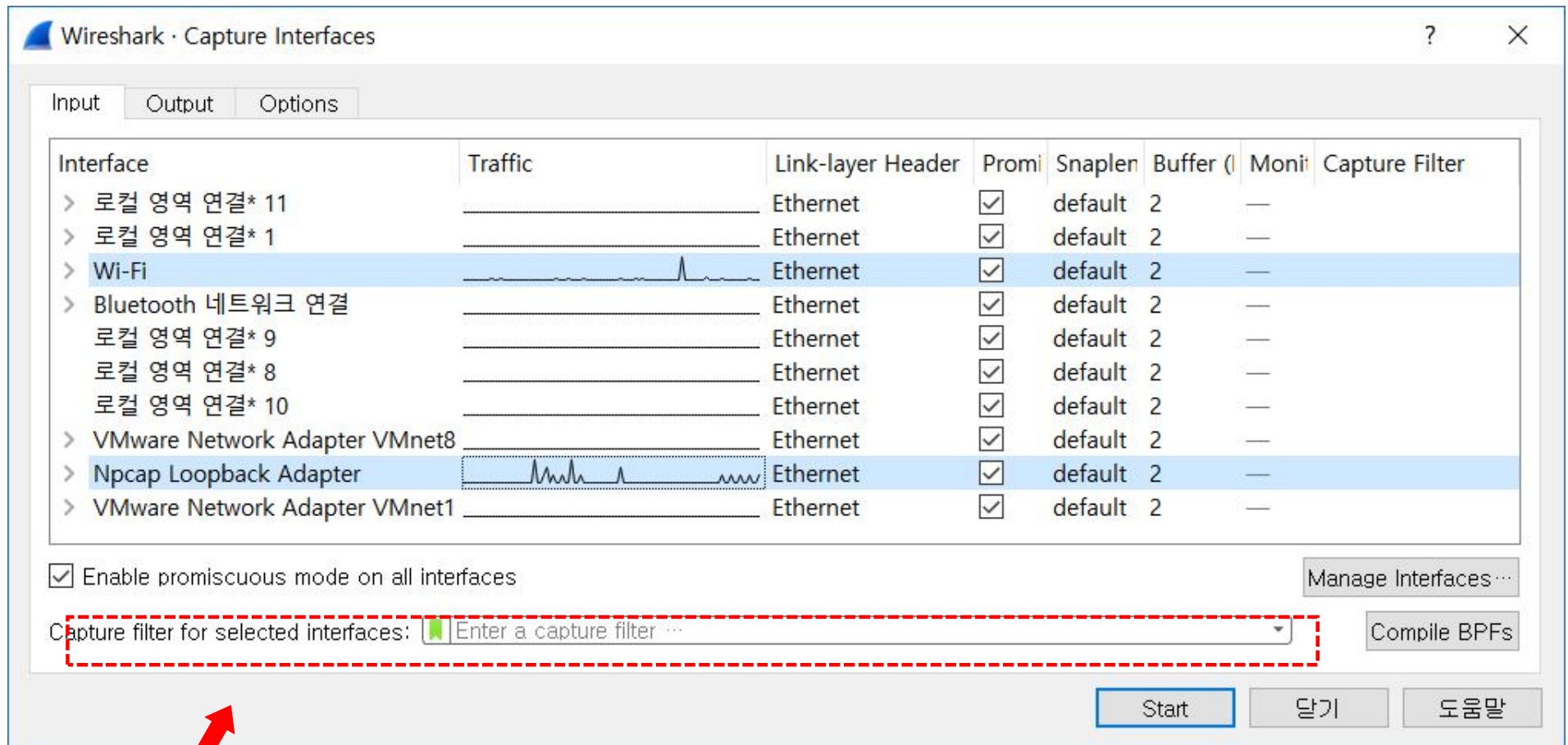
2.2 Wireshark Filter

- 캡처 필터(Capture Filter)
- 디스플레이 필터(Display Filter)
- Conversation 필터
- 스트림 따라가기(Stream Follow)

- 캡처 필터(Capture Filter)
 - 패킷이 캡처 될 때 지정
 - 지정된 표현식에 포함/제외된 패킷만 캡처
- 디스플레이 필터(Display Filter)
 - 원하지 않는 패킷을 숨김
 - 지정된 표현식을 기반으로 원하는 패킷을 보기

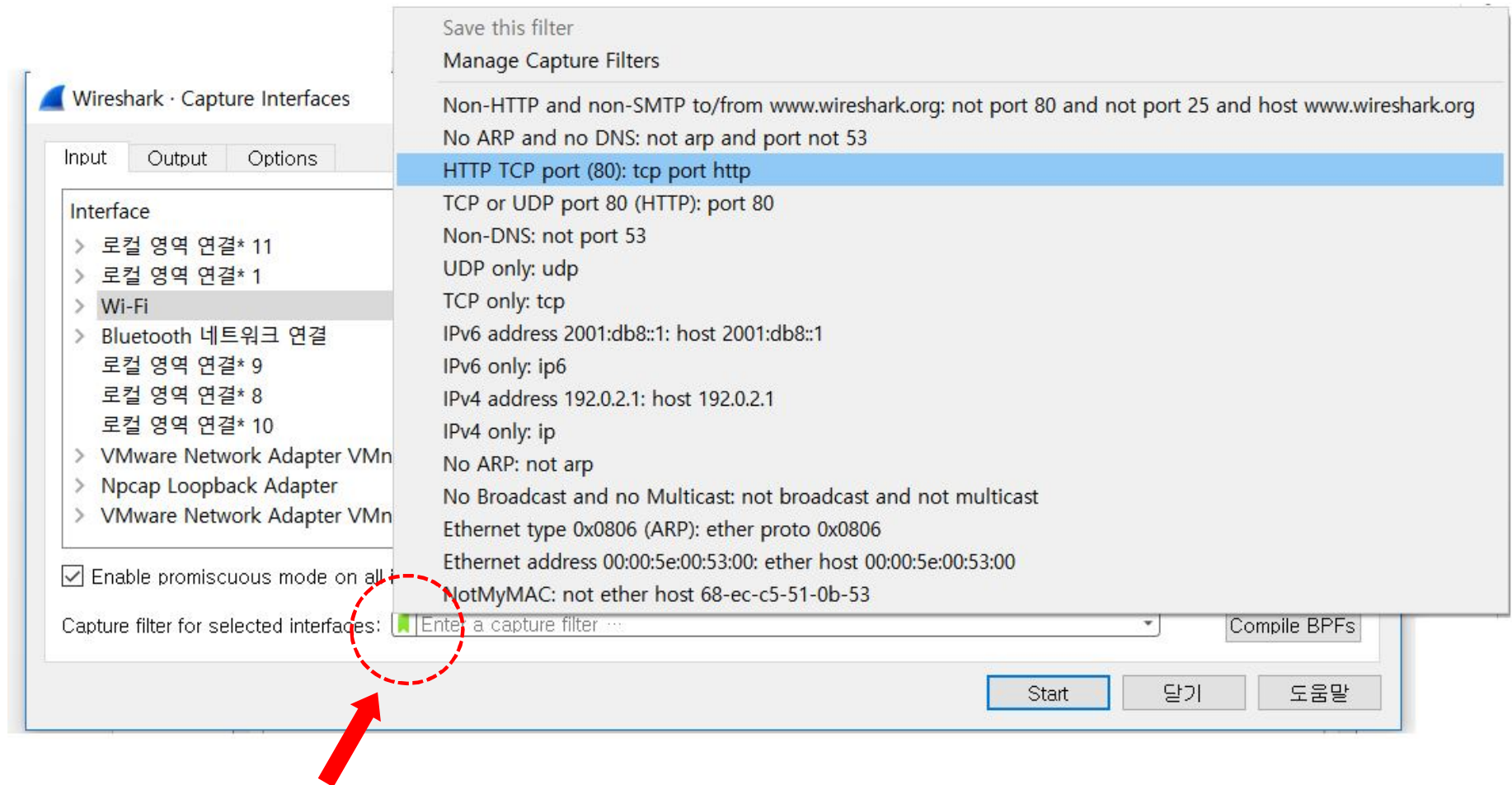
1) Capture Filter

- Capture Option 창에서 수집 필터 적용



Capture Option 창에서 수집 필터 적용

❶ Capture Option > 수집 필터 체크표기 화살표



수집 필터 체크표기 화살표

주소 기반의 트래픽 수집

❶ 특정 IP 주소에서/로 오는 트래픽 수집

- host 10.3.1.1
- host 2406:da00:ff00::6b16:f02d
- not host 10.3.1.1
- src host 10.3.1.1
- dst host 10.3.1.1
- host 10.3.1.1 or host 10.3.1.2
- host www.espn.com

주소 기반의 트래픽 수집

② IP 주소 범위에서/로 오는 트래픽 수집

- net 10.3.0.0/16
- net 10.3.0.0 mask 255.255.0.0
- ipv6 net 2406:da00:ff00::/64
- not dst net 10.3.0.0/16
- dst net 10.3.0.0/16
- src net 10.3.0.0/16

주소 기반의 트래픽 수집

③ 브로드캐스트 또는 멀티캐스트 트래픽 수집

- ip broadcast
- ip multicast
- dst host ff02::1
- dst host ff02::2

주소 기반의 트래픽 수집

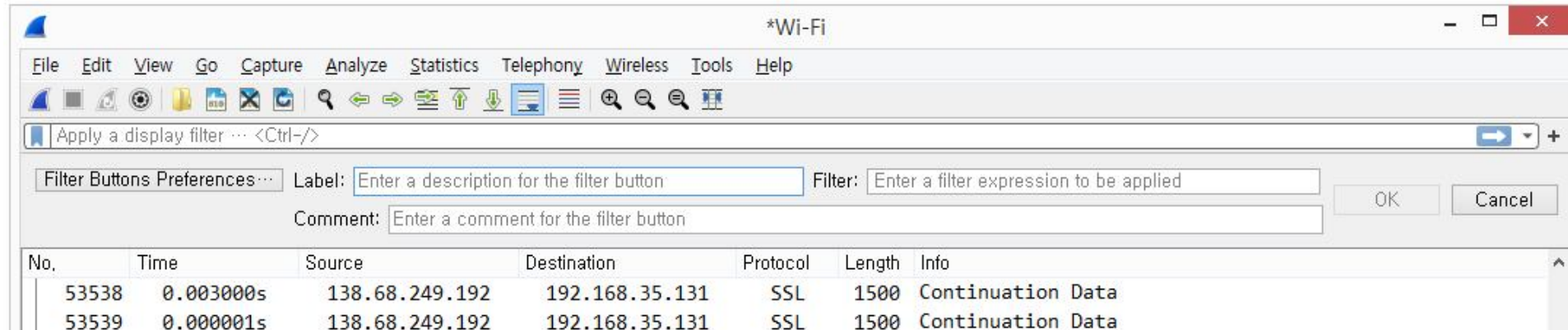
④ MAC 주소 기반의 트래픽 수집

- ether host 00:08:15:00:08:15
- ether src 00:08:15:00:08:15
- ether dst 00:08:15:00:08:15
- not ether host 00:08:15:00:08:15

특정 애플리케이션에 대한 트래픽 수집

- port 53
- not port 53
- port 80
- udp port 67
- tcp port 21
- portrange 1-80
- tcp portrange 1-80
- port 20 or port 21
- host 10.3.1.1 and port 80
- host 10.3.1.1 and not port 80
- udp src port 68 and udp dst port 67

2) 디스플레이 필터



적절한 디스플레이 필터 문법 사용

- 간단한 디스플레이 필터 문법

arp

ip

ipv6

tcp

적절한 디스플레이 필터 문법 사용

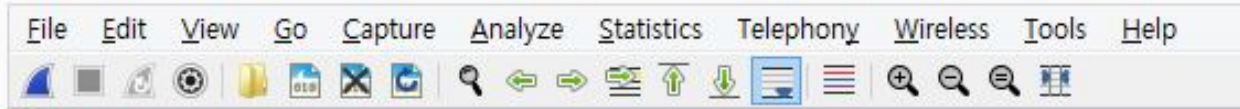
<<디스플레이 필터 오류 탐지 메커니즘 >>

- 대소문자 구분

- 적색 배경

- 문법 검사 실패

- 동작하지 않음



The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The filter bar at the top of the packet list shows 'http.re' with a red background, indicating a syntax error. The packet list below shows two packets:

No.	Time	Source	Destination	Protocol
1	0.000000s	138.68.249.192	192.168.35.131	SSL
2	0.000069s	192.168.35.131	138.68.249.192	TCP

- 녹색 배경

- 문법 이상 없음

- ‘논리 검사’는 하지 않음 (예) http && udp

- 황색 배경

- 필터가 원하는 대로 동작하지 않는 것을 경고 (예) ip.addr != 10.1.1.1

디스플레이필터와 연산자 비교

연산자	영어표기	예제
==	eq	ip.src == 10.2.2.2
!=	ne	tcp.srcport != 80
>	gt	frame.time_relative > 1
<	lt	tcp.window_size < 1460
>=	ge	dns.count.answers >=10
<=	lt	ip.ttl < 10
	contains	http contain "GET"

캡처 필터 vs 디스플레이 필터

캡처 필터 구문 예제	디스플레이 필터 예제
host 172.16.1.1	ip.host == 172.16.1.1
src host 172.16.1.1	ip.src ==172.16.1.1
dst host 172.16.1.1	ip.dst ==172.16.1.1
port 8080	tcp.port == 8080
!port 8080	!tcp.port = 8080

❶ 단순 IP 주소 호스트에게/부터의 트래픽 필터링

- `ip.addr == 10.3.1.1`
- `!ip.addr == 10.3.1.1`
- `ipv6.addr == 2406:da00:ff00::6b16:f02d`
- `ip.src == 10.3.1.1`
- `ip.dst == 10.3.1.1`
- `ip.host == www.wireshark.org`

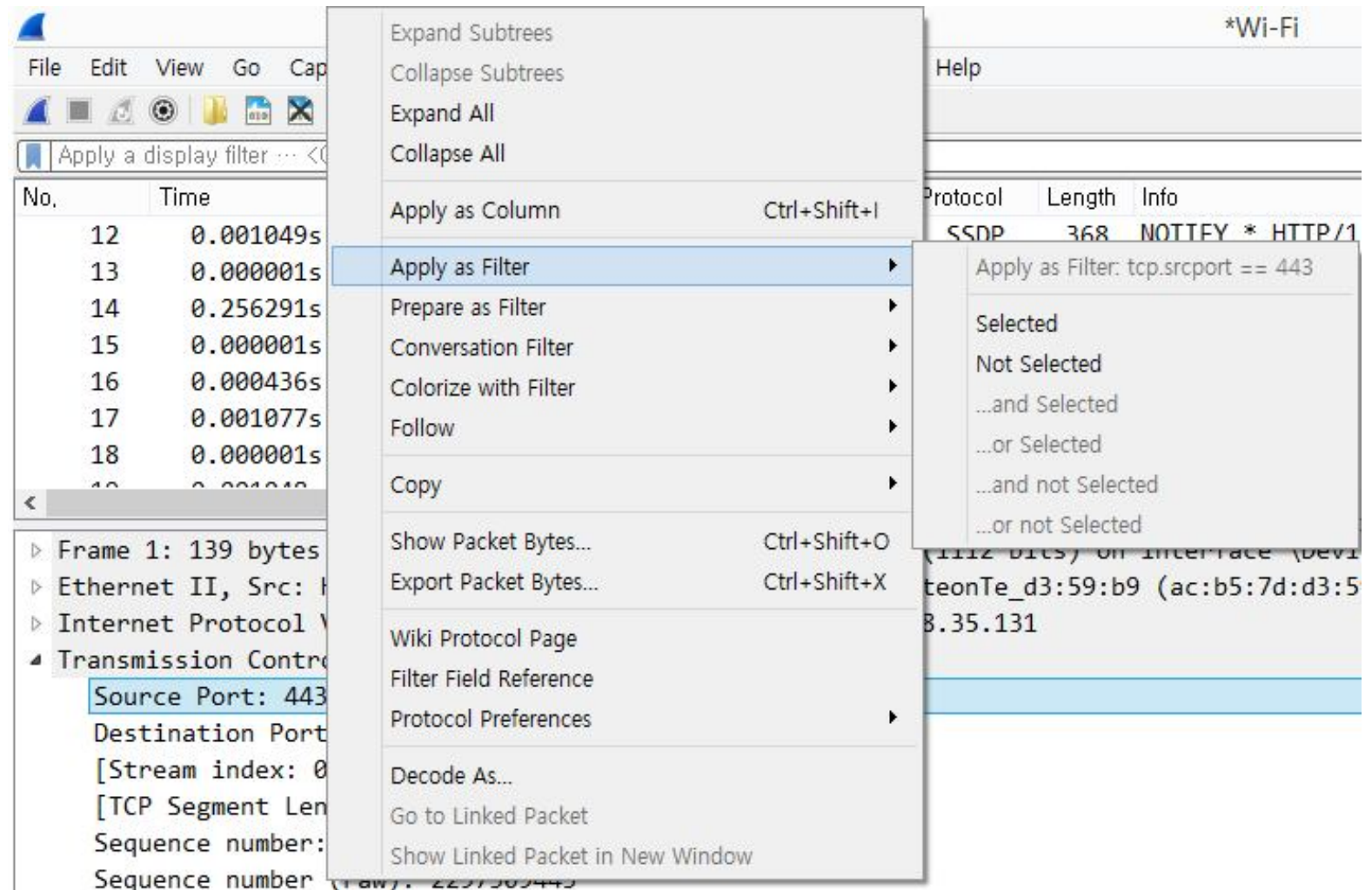
② 주소 범위에게/부터의 트래픽 필터링

- `ip.addr > 10.3.0.1 && ip.addr < 10.3.0.5`
- `(ip.addr >= 10.3.0.1 && ip.addr <= 10.3.0.6) && !ip.addr == 10.3.0.3`
- `ipv6.addr == fe80:: && ipv6.addr < fec0::`

③ IP 서브넷에서/으로부터 트래픽 필터링

- `ip.addr == 10.3.0.0/16`
- `ip.addr == 10.3.0.0/16 && !ip.addr == 10.3.0.3`
- `!ip.addr == 10.3.0.0/16 && !ip.addr == 10.2.0.0/16`

- Apply as Filter
- Prepare as Filter



Selected

$A == B$

Not Selected

$!(A == B)$

... and Selected

$(A == B) \&\& (C \dots)$

... or Selected

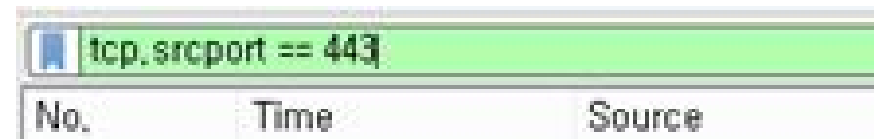
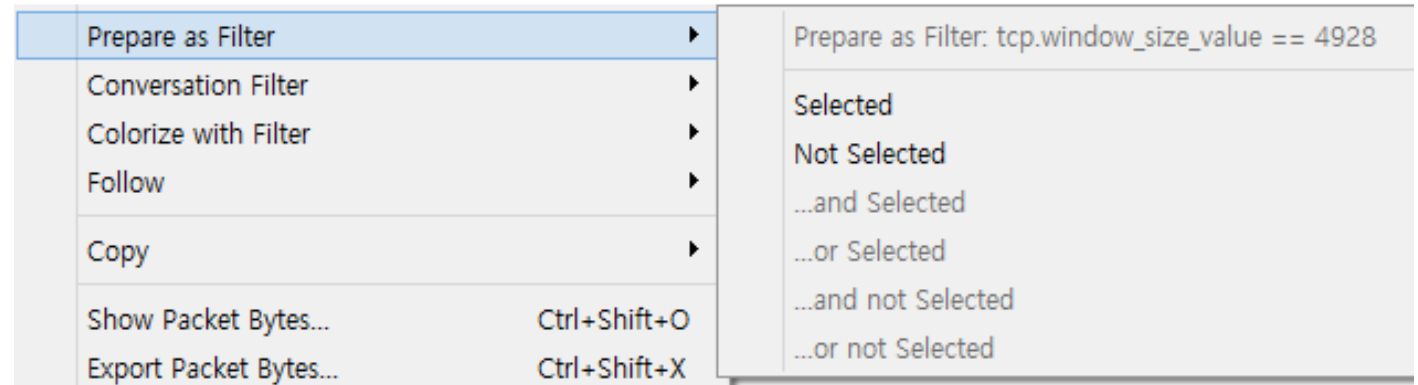
$(A == B) \parallel (C \dots)$

... and Not Selected

$(A == B) \&\& !(C \dots)$

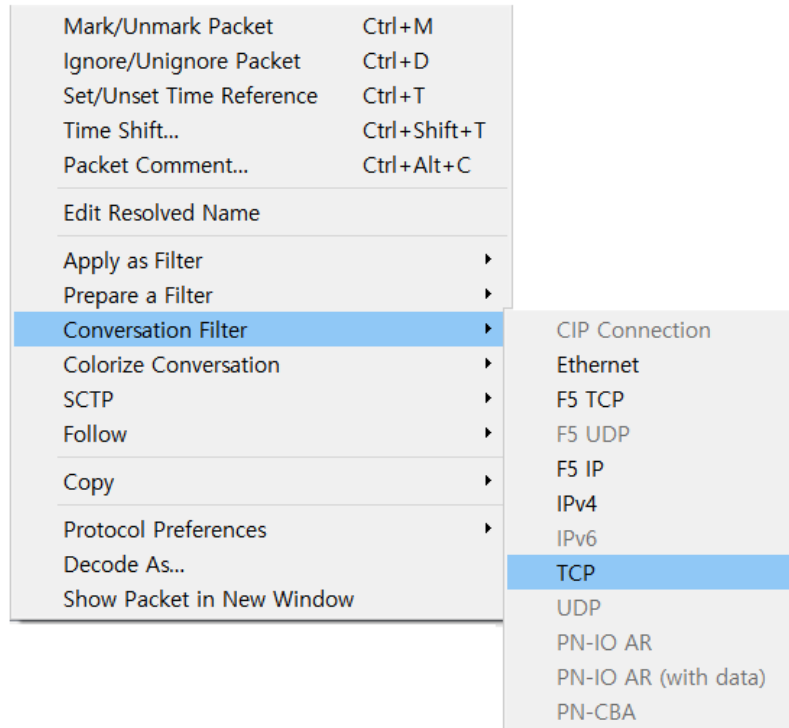
... or Not Selected

$(A == B) \parallel !(C \dots)$



3) Conversation Filter

❶ 패킷 리스트 > 패킷선택 > 오른쪽 마우스 클릭 > Conversation Filter > TCP



추적파일 : <http-esp101.pcapng>

(ip.addr eq 24.6.173.220 and ip.addr eq 199.181.132.250) and (tcp.port eq 19941 and tcp.port eq 80)						
No.	Time	Source	Destination	Protocol	Length	Info
5	0.000000	24.6.173.220	199.181.132.250	TCP	66	19941 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=
6	0.031335	199.181.132.250	24.6.173.220	TCP	66	80 → 19941 [SYN, ACK] Seq=0 Ack=1 Win=4380
7	0.000126	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=1 Ack=1 Win=65700 Len
8	0.000665	24.6.173.220	199.181.132.250	HTTP	603	GET / HTTP/1.1
9	0.041099	199.181.132.250	24.6.173.220	HTTP	484	HTTP/1.1 301 Moved Permanently (text/html
31	0.199860	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=550 Ack=431 Win=65268
4891	68.873340	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [RST, ACK] Seq=550 Ack=431 Win=

② Statistics > Conversation Filter

추적파일 : http-espn101.pcapng

Wireshark · Conversations · http-espn101.pcapng

Ethernet · 1IPv4 · 37IPv6TCP · 63UDP · 82

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A	
24.6.173.220	75.75.75.75	180	22 k	90	6973	90	15 k	0.000000	21.8143	2557		5526
24.6.173.220	199.181.132.250	7	1381	5	831	2	550	0.030245	69.1464	96		63
24.6.173.220	68.71.216.176	127	134 k	38	7147	89	127 k	0.168701	24.5121	2332		41 k
24.6.173.220	184.84.222.48	720	649 k	265	43 k	455	605 k	0.322923	70.0159	5024		69 k
24.6.173.220	143.127.102.125	10	1229	5	514	5	715	0.377829	0.1381	29 k		41 k
24.6.173.220	70.42.13.100	12	2578	7	1903	5	675	2.433476	14.8802	1023		362
24.6.173.220	68.71.212.151	7	1293	5	828	2	465	2.437970	66.7377	99		55
24.6.173.220	74.125.224.59	142	115 k	51	9643	91	105 k	2.843065	66.3320	1162		12 k
24.6.173.220	184.84.222.152	303	286 k	110	25 k	193	261 k	3.261301	70.9168	2865		29 k

Wireshark · Conversations · http-espn101.pcapng

Ethernet · 1IPv4 · 37IPv6TCP · 63UDP · 82

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
24.6.173.220	143.127.102.125	10	1229	5	514	5	715	0.377829	0.1381	29 k	41 k
24.6.173.220	68.71.212.151	7	1293								
24.6.173.220	199.181.132.250	7	1381								
24.6.173.220	107.20.148.253	10	1400								
24.6.173.220	184.84.222.64	8	1422								
24.6.173.220	184.51.159.181	10	1437	6	888						
24.6.173.220	184.84.183.147	8	1573	5	699						
24.6.173.220	107.22.175.32	10	1602	6	1068	4	55				
24.6.173.220	184.84.222.112	8	2120	5	701	3	141				
24.6.173.220	64.95.73.7	9	2311	6	1304	3	100				
24.6.173.220	68.71.220.175	7	2355	5	1171	2	118				

② Statistics > Conversation Filter

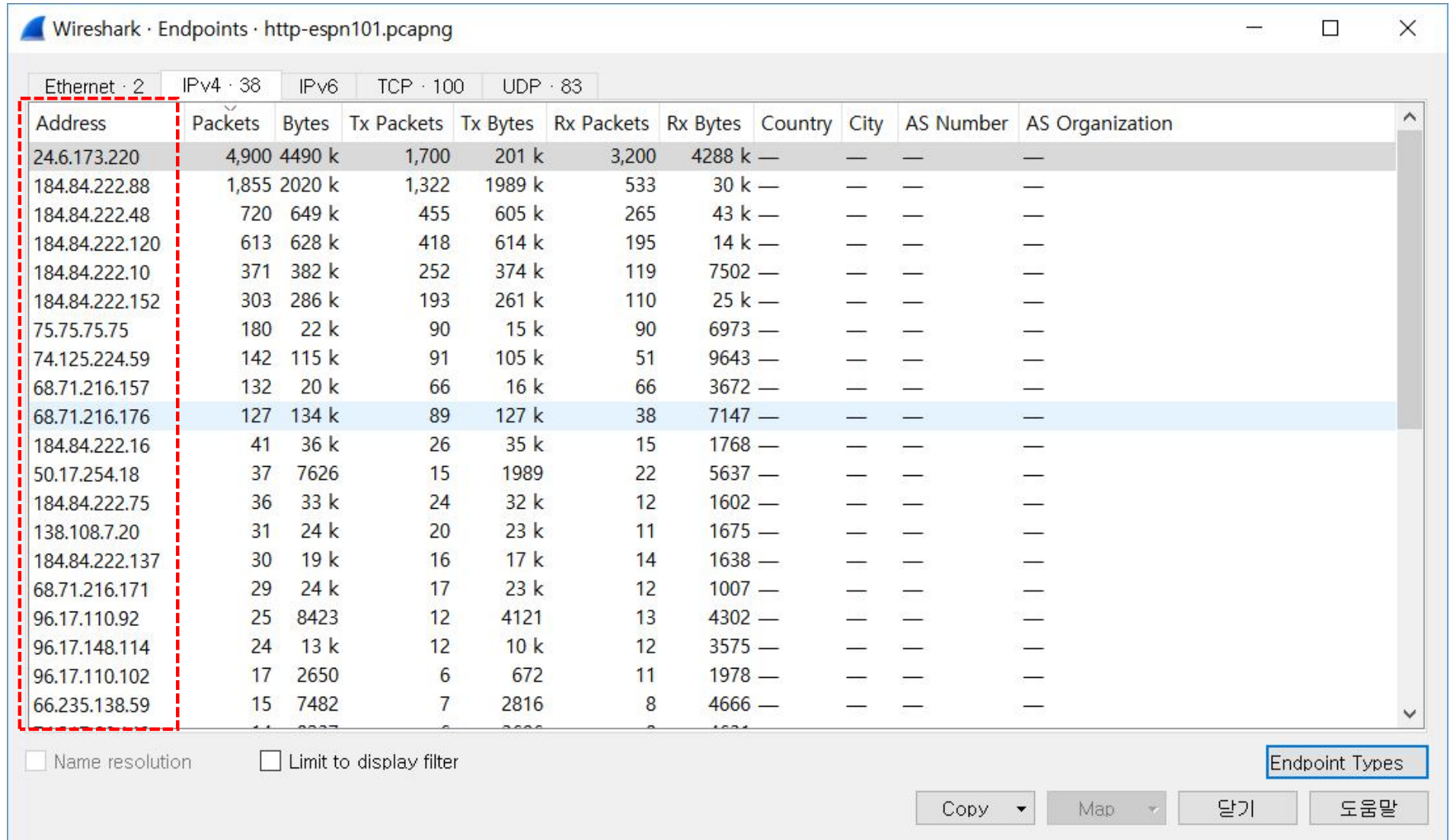
- Packets 필드를 기준으로 내림 차순으로 정렬
- 첫 번째 패킷 선택
- 오른쪽 마우스 클릭 > Apply as Filter > Selected > A → B

Wireshark · Conversations · http-espn101.pcapng

Ethernet · 1		IPv4 · 37		IPv6	TCP · 63		UDP · 82					
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B		
24.6.173.220	184.84.222.88	1,855	2020 k	533						3765		
24.6.173.220	184.84.222.48	720	649 k	265						5024		
24.6.173.220	184.84.222.120	613	628 k	195						1882		
24.6.173.220	184.84.222.10	371	382 k	119						980		
24.6.173.220	184.84.222.152	303	286 k	110						2865		
24.6.173.220	75.75.75.75	180	22 k	90	6973	90				2557		
24.6.173.220	74.125.224.59	142	115 k	51	9643	91				1162		
24.6.173.220	68.71.216.157	132	20 k	66	3672	66	16 k	21.802866	4	658		
24.6.173.220	68.71.216.176	127	134 k	38	7147	89	127 k	0.168701	2	2332		
24.6.173.220	184.84.222.16	41	36 k	15	1768	26	35 k	7.951909	6	231		
24.6.173.220	50.17.254.18	37	7626	22	5637	15	1989	9.581595	2.8209	15 k		
24.6.173.220	184.84.222.75	36	33 k	12	1602	24	32 k	5.377013	63.7964	200		

[참고] Endpoints

Statistics > Endpoints



Wireshark · Endpoints · http-espn101.pcapng

Ethernet · 2 IPv4 · 38 IPv6 TCP · 100 UDP · 83

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	AS Number	AS Organization
24.6.173.220	4,900	4490 k	1,700	201 k	3,200	4288 k	—	—	—	—
184.84.222.88	1,855	2020 k	1,322	1989 k	533	30 k	—	—	—	—
184.84.222.48	720	649 k	455	605 k	265	43 k	—	—	—	—
184.84.222.120	613	628 k	418	614 k	195	14 k	—	—	—	—
184.84.222.10	371	382 k	252	374 k	119	7502	—	—	—	—
184.84.222.152	303	286 k	193	261 k	110	25 k	—	—	—	—
75.75.75.75	180	22 k	90	15 k	90	6973	—	—	—	—
74.125.224.59	142	115 k	91	105 k	51	9643	—	—	—	—
68.71.216.157	132	20 k	66	16 k	66	3672	—	—	—	—
68.71.216.176	127	134 k	89	127 k	38	7147	—	—	—	—
184.84.222.16	41	36 k	26	35 k	15	1768	—	—	—	—
50.17.254.18	37	7626	15	1989	22	5637	—	—	—	—
184.84.222.75	36	33 k	24	32 k	12	1602	—	—	—	—
138.108.7.20	31	24 k	20	23 k	11	1675	—	—	—	—
184.84.222.137	30	19 k	16	17 k	14	1638	—	—	—	—
68.71.216.171	29	24 k	17	23 k	12	1007	—	—	—	—
96.17.110.92	25	8423	12	4121	13	4302	—	—	—	—
96.17.148.114	24	13 k	12	10 k	12	3575	—	—	—	—
96.17.110.102	17	2650	6	672	11	1978	—	—	—	—
66.235.138.59	15	7482	7	2816	8	4666	—	—	—	—

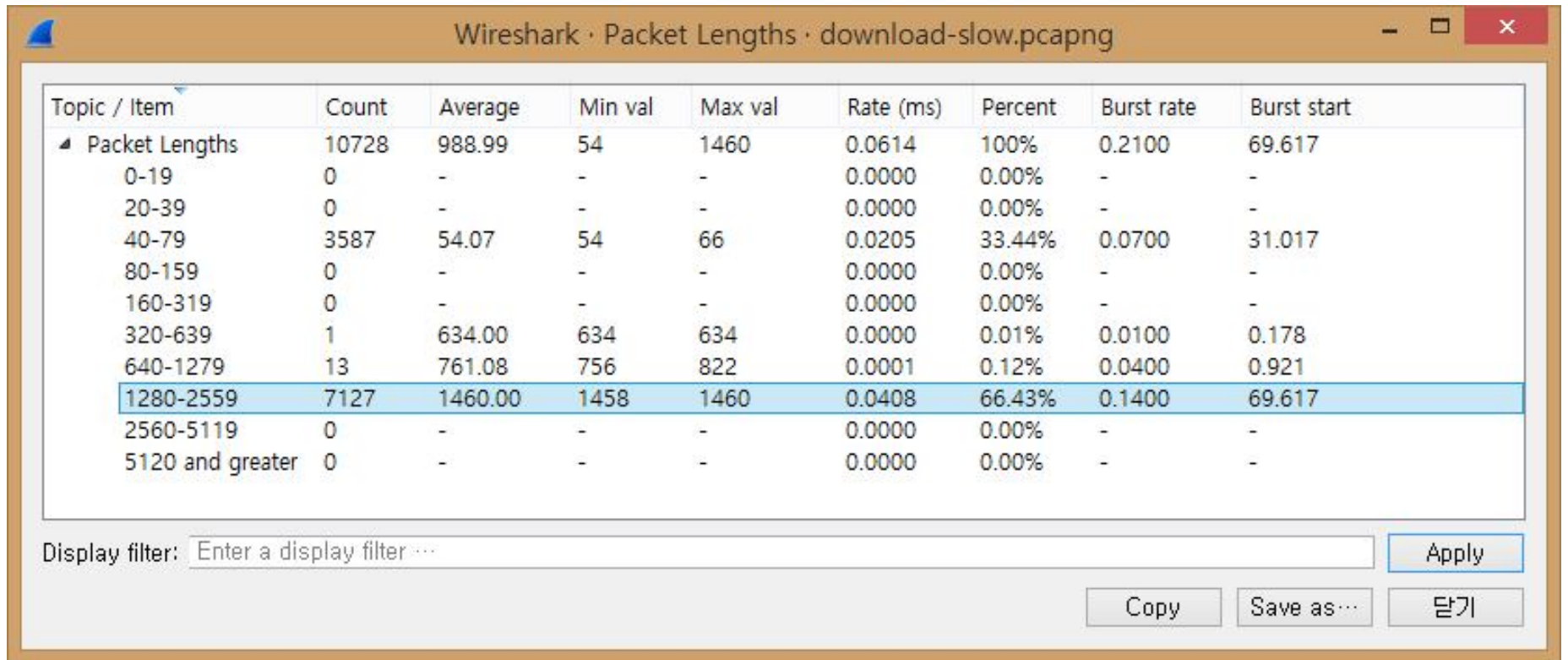
☐ Name resolution ☐ Limit to display filter

Endpoint Types

Copy Map 달기 도움말

* Statistics > Packet Lengths

- 대용량 캡처 파일의 구성을 이해하고자 할 때 유용
- 길이에 따라 패킷 분포를 확인

The image shows the 'Wireshark · Packet Lengths · download-slow.pcapng' window. It contains a table with statistics for various packet length ranges. The '1280-2559' range is highlighted in blue, showing a count of 7127 and a percentage of 66.43%. Below the table is a 'Display filter' input field and buttons for 'Apply', 'Copy', 'Save as...', and '닫기' (Close).

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
▲ Packet Lengths	10728	988.99	54	1460	0.0614	100%	0.2100	69.617
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	3587	54.07	54	66	0.0205	33.44%	0.0700	31.017
80-159	0	-	-	-	0.0000	0.00%	-	-
160-319	0	-	-	-	0.0000	0.00%	-	-
320-639	1	634.00	634	634	0.0000	0.01%	0.0100	0.178
640-1279	13	761.08	756	822	0.0001	0.12%	0.0400	0.921
1280-2559	7127	1460.00	1458	1460	0.0408	66.43%	0.1400	69.617
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

- 길이가 긴 패킷이 66.43% 차지, 짧은 길의 패킷은 33.44%

*Statistics > Protocol Hierarchy

Wireshark · Protocol Hierarchy Statistics · http_espn.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
▼ Frame	100.0	956	100.0	652181	2548 k	0	0	0
▼ Ethernet	100.0	956	2.1	13384	52 k	0	0	0
▼ Internet Protocol Version 4	100.0	956	2.9	19120	74 k	0	0	0
▼ User Datagram Protocol	2.9	28	0.0	224	875	0	0	0
Domain Name System	2.9	28	0.3	2017	7880	28	2017	7880
▼ Transmission Control Protocol	97.1	928	94.7	617436	2412 k	426	14032	54 k
▼ Hypertext Transfer Protocol	52.5	502	90.1	587338	2294 k	441	517225	2020 k
Unreassembled Fragmented Packet	0.1	1	0.0	0	0	1	0	0
▼ Portable Network Graphics	1.5	14	1.3	8607	33 k	8	3064	11 k
Unreassembled Fragmented Packet	0.6	6	0.0	0	0	6	0	0
Media Type	0.3	3	0.4	2741	10 k	3	2741	10 k
Line-based text data	1.5	14	2.3	15259	59 k	14	15259	59 k
▼ JPEG File Interchange Format	1.9	18	2.5	16201	63 k	13	11575	45 k
Unreassembled Fragmented Packet	0.5	5	0.0	0	0	5	0	0
eXtensible Markup Language	0.2	2	0.9	5868	22 k	2	5868	22 k
▼ Compuserve GIF	0.9	9	0.8	4946	19 k	5	1254	4899
Unreassembled Fragmented Packet	0.4	4	0.0	0	0	4	0	0

• Statistics > Conversation

Wireshark · Conversations · http_espn_fail.pcapng

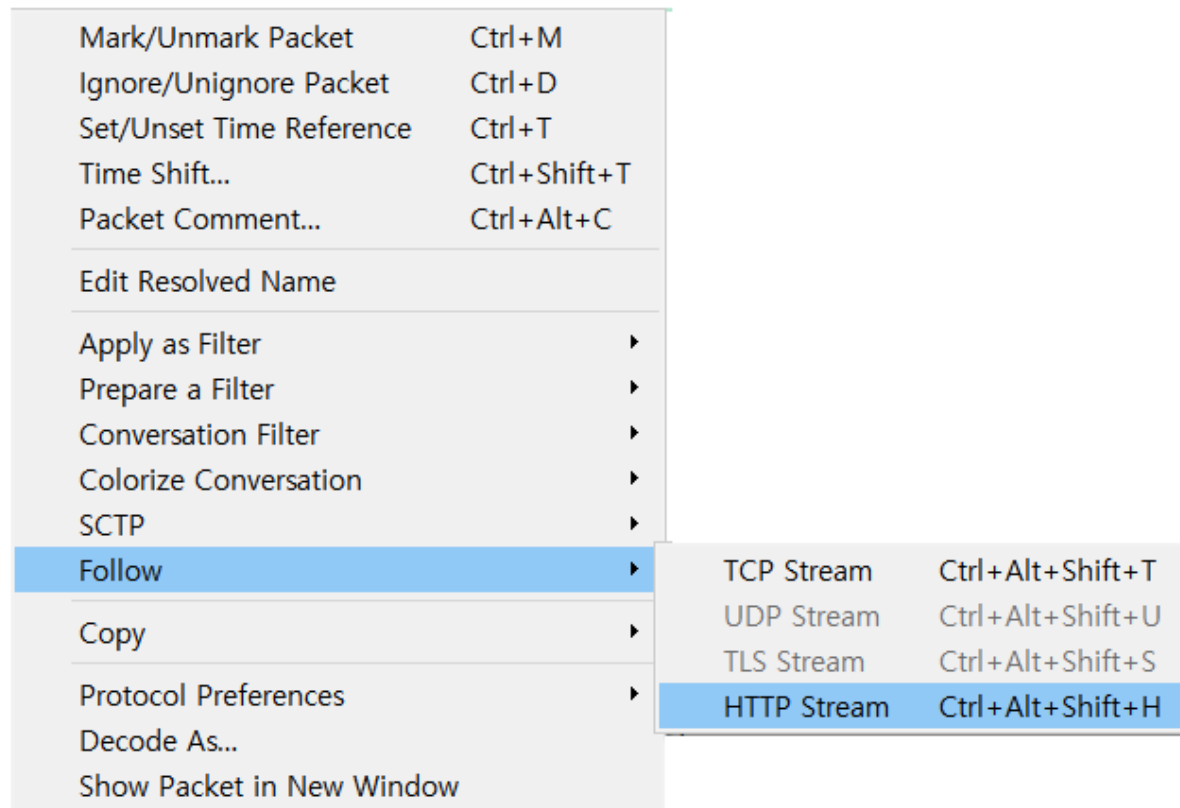
Ethernet · 1 IPv4 · 8 IPv6 TCP · 16 UDP · 7											
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
172.16.16.154	4.2.2.1	14	1627	7	521	7	1106	0.000000	0.6639	6278	
172.16.16.154	68.71.212.158	13	2032	7	832	6	1200	0.027167	90.8752	73	
172.16.16.154	199.181.133.61	61	49 k	24	1953	37	47 k	0.238547	91.0836	171	
172.16.16.154	203.0.113.94	93	6774	93	6774	0	0	0.430071	94.5936	572	
172.16.16.154	72.21.91.8	92	70 k	43	3170	49	67 k	0.526867	60.5532	418	
172.16.16.154	72.246.56.35	247	196 k	113	8315	134	188 k	0.527902	90.8063	732	
172.16.16.154	69.31.75.194	19	9949	9	1007	10	8942	0.579477	90.6593	88	
172.16.16.154	72.246.56.83	30	20 k	15	1518	15	19 k	0.659868	45.3449	267	

4) 스트림 따라가기(Stream Follow)

- 여러 패킷의 데이터를 통합해 쉽게 읽을 수 있는 형식으로 재구성 (재조립)
- 4가지 유형의 스트림
 - TCP stream
 - UDP Stream
 - SSL Stream
 - HTTP Stream

* 단일 TCP나 UDP 대화 필터링

TCP 또는 HTTP 패킷 선택 > 오른쪽 마우스 클릭 > Follow > HTTP Stream



추적파일 : http-espn101.pcapng

Wireshark · Follow HTTP Stream (tcp.stream eq 0) · http-espn101.pcapng

```
GET / HTTP/1.1
Accept: application/x-ms-application, image/jpeg, application/xaml+xml, image/gif, image/pjpeg,
application/x-ms-xbap, application/vnd.ms-excel, application/vnd.ms-powerpoint, application/
msword, */*
Accept-Language: en-US
User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; WOW64; Trident/4.0; GTB7.2; SLCC2;
.NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; HPDTEF;
.NET4.0C; InfoPath.3; MS-RTC LM 8; BRI/2)
Accept-Encoding: gzip, deflate
Host: www.espn.com
Connection: Keep-Alive

HTTP/1.1 301 Moved Permanently
Date: Sat, 07 Jan 2012 21:59:44 GMT
Server: Apache
Location: http://espn.go.com/
Content-Length: 227
X-Cnection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="http://espn.go.com/">here</a>.</p>
</body></html>
```

client pkt(s), server pkt(s), turn(s)

Entire conversation (979 bytes)

Show and save data as ASCII

Find: Find Next

Filter Out This Stream Print Save as... Back 닫기 도움말

http-espn101.pcapng

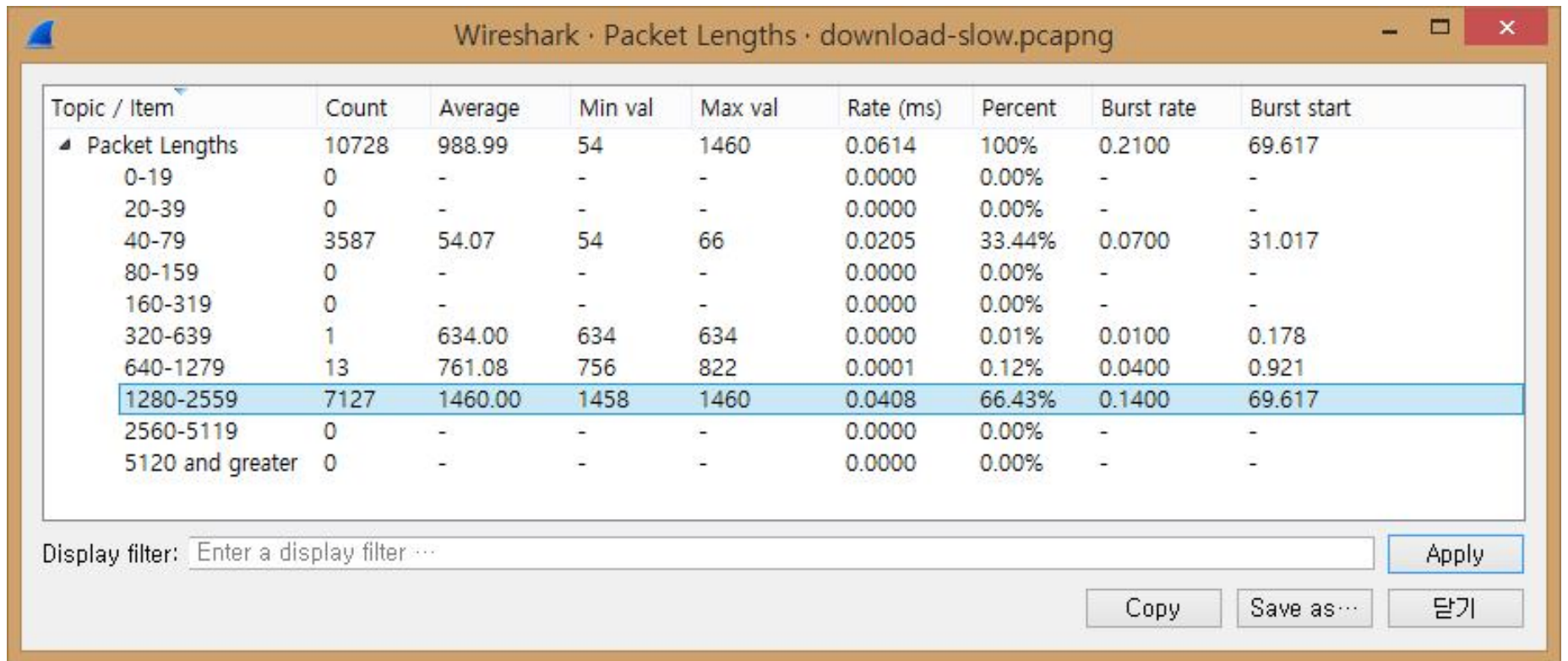
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 0

No.	Time	Source	Destination	Protocol	Length	Info
5	0.000000	24.6.173.220	199.181.132.250	TCP	66	19941 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=
6	0.031335	199.181.132.250	24.6.173.220	TCP	66	80 → 19941 [SYN, ACK] Seq=0 Ack=1 Win=4380 Len=0 M
7	0.000126	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=1 Ack=1 Win=65700 Len=0
8	0.000665	24.6.173.220	199.181.132.250	HTTP	603	GET / HTTP/1.1
9	0.041099	199.181.132.250	24.6.173.220	HTTP	484	HTTP/1.1 301 Moved Permanently (text/html)
31	0.199860	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=550 Ack=431 Win=65268 Len=0
4891	68.873340	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [RST, ACK] Seq=550 Ack=431 Win=0 Len=0

* Statistics > Packet Lengths

- 대용량 캡처 파일의 구성을 이해하고자 할 때 유용
- 길이에 따라 패킷 분포를 확인

The image shows the 'Wireshark · Packet Lengths · download-slow.pcapng' window. It contains a table with 9 columns: Topic / Item, Count, Average, Min val, Max val, Rate (ms), Percent, Burst rate, and Burst start. The 'Packet Lengths' section is expanded, showing various size ranges. The '1280-2559' range is highlighted in blue, indicating it is the selected item. Below the table is a 'Display filter' input field and buttons for 'Apply', 'Copy', 'Save as...', and '닫기' (Close).

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
Packet Lengths	10728	988.99	54	1460	0.0614	100%	0.2100	69.617
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	3587	54.07	54	66	0.0205	33.44%	0.0700	31.017
80-159	0	-	-	-	0.0000	0.00%	-	-
160-319	0	-	-	-	0.0000	0.00%	-	-
320-639	1	634.00	634	634	0.0000	0.01%	0.0100	0.178
640-1279	13	761.08	756	822	0.0001	0.12%	0.0400	0.921
1280-2559	7127	1460.00	1458	1460	0.0408	66.43%	0.1400	69.617
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

- 길이가 긴 패킷이 66.43% 차지, 짧은 길의 패킷은 33.44%

*Statistics > Protocol Hierarchy

Wireshark - Protocol Hierarchy Statistics - http_espn.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
▼ Frame	100.0	956	100.0	652181	2548 k	0	0	0
▼ Ethernet	100.0	956	2.1	13384	52 k	0	0	0
▼ Internet Protocol Version 4	100.0	956	2.9	19120	74 k	0	0	0
▼ User Datagram Protocol	2.9	28	0.0	224	875	0	0	0
Domain Name System	2.9	28	0.3	2017	7880	28	2017	7880
▼ Transmission Control Protocol	97.1	928	94.7	617436	2412 k	426	14032	54 k
▼ Hypertext Transfer Protocol	52.5	502	90.1	587338	2294 k	441	517225	2020 k
Unreassembled Fragmented Packet	0.1	1	0.0	0	0	1	0	0
▼ Portable Network Graphics	1.5	14	1.3	8607	33 k	8	3064	11 k
Unreassembled Fragmented Packet	0.6	6	0.0	0	0	6	0	0
Media Type	0.3	3	0.4	2741	10 k	3	2741	10 k
Line-based text data	1.5	14	2.3	15259	59 k	14	15259	59 k
▼ JPEG File Interchange Format	1.9	18	2.5	16201	63 k	13	11575	45 k
Unreassembled Fragmented Packet	0.5	5	0.0	0	0	5	0	0
eXtensible Markup Language	0.2	2	0.9	5868	22 k	2	5868	22 k
▼ Compuserve GIF	0.9	9	0.8	4946	19 k	5	1254	4899
Unreassembled Fragmented Packet	0.4	4	0.0	0	0	4	0	0

• Statistics > Conversation

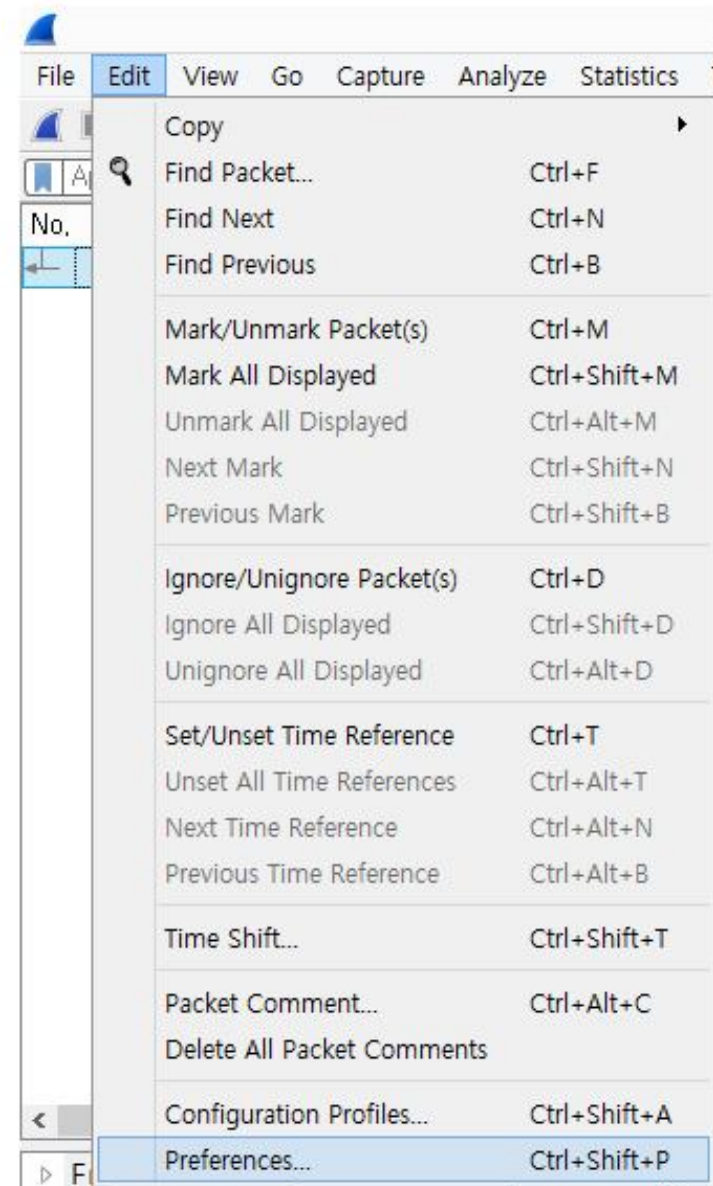
Wireshark - Conversations - http_espn_fail.pcapng

Ethernet · 1 IPv4 · 8 IPv6 TCP · 16 UDP · 7											
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
172.16.16.154	4.2.2.1	14	1627	7	521	7	1106	0.000000	0.6639	6278	
172.16.16.154	68.71.212.158	13	2032	7	832	6	1200	0.027167	90.8752	73	
172.16.16.154	199.181.133.61	61	49 k	24	1953	37	47 k	0.238547	91.0836	171	
172.16.16.154	203.0.113.94	93	6774	93	6774	0	0	0.430071	94.5936	572	
172.16.16.154	72.21.91.8	92	70 k	43	3170	49	67 k	0.526867	60.5532	418	
172.16.16.154	72.246.56.35	247	196 k	113	8315	134	188 k	0.527902	90.8063	732	
172.16.16.154	69.31.75.194	19	9949	9	1007	10	8942	0.579477	90.6593	88	
172.16.16.154	72.246.56.83	30	20 k	15	1518	15	19 k	0.659868	45.3449	267	

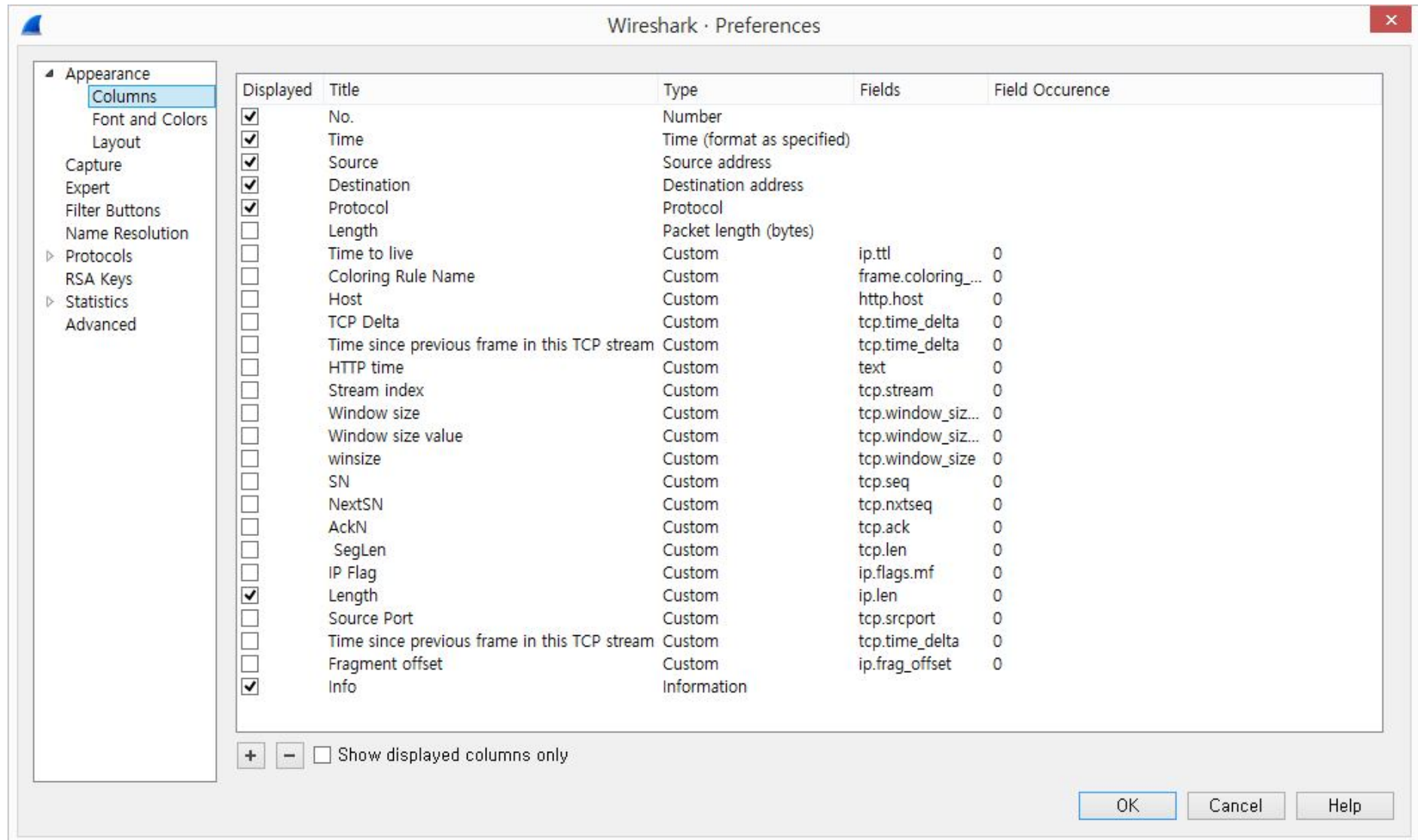
2.3 와이어샤크 뷰와 설정 맞춤화

1) 패킷 목록 창에 칼럼(열) 추가

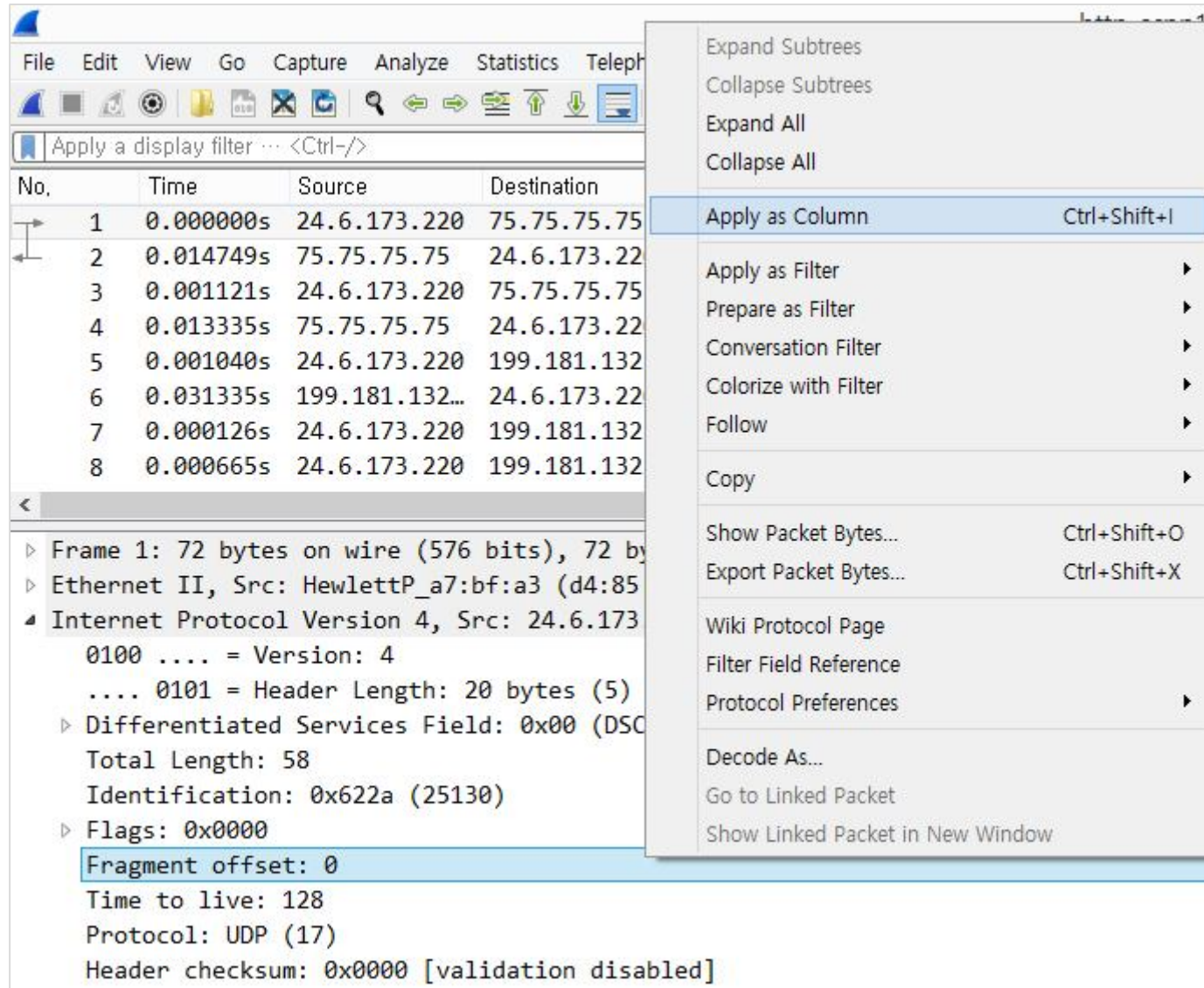
1 Edit > Preference



Edit > Preference > Appearance > Columns



1) 패킷 목록 창에 칼럼(열) 추가 ②



No.	Time	Source	Destination
4568	0.000069s	24.6.173.220	184.84.222.88
4572	0.000061s	24.6.173.220	184.84.222.88
4575	0.000065s	24.6.173.220	184.84.222.88
4579	0.000063s	24.6.173.220	184.84.222.88
4582	0.000072s	24.6.173.220	184.84.222.88
4585	0.000064s	24.6.173.220	184.84.222.88
4591	0.000119s	24.6.173.220	184.84.222.88
4895	0.000313s	24.6.173.220	184.84.222.88
1345	0.000847s	24.6.173.220	184.84.222.97
1349	0.000093s	24.6.173.220	184.84.222.97
1350	0.000503s	24.6.173.220	184.84.222.97
1354	0.000268s	24.6.173.220	184.84.222.97
1357	0.000378s	24.6.173.220	184.84.222.97
4870	0.000189s	24.6.173.220	184.84.222.97
5	0.001040s	24.6.173.220	199.181.132.250
7	0.000126s	24.6.173.220	199.181.132.250
8	0.000665s	24.6.173.220	199.181.132.250
31	0.000574s	24.6.173.220	199.181.132.250
4891	0.000240s	24.6.173.220	199.181.132.250
2	0.014749s	75.75.75.75	24.6.173.220

▶ Frame 2: 88 bytes on wire (704 bits), 88 bytes captured
 ▶ Ethernet II, Src: Cadant_31:bb:c1 (00:01:5c:31:bb:c1)
 ▶ Internet Protocol Version 4, Src: 75.75.75.75, Dst: 24.6.173.220

0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 ▶ Differentiated Services Field: 0x40 (DSCP: CS2, ECN: Not-ECT)
 Total Length: 74
 Identification: 0x0000 (0)
 ▶ Flags: 0x4000, Don't fragment
 Fragment offset: 0
 Time to live: 59
 Protocol: UDP (17)
 Header checksum: 0xe2ea [validation disabled]
 [Header checksum status: Unverified]
 Source: 75.75.75.75

0000	d4 85 64 a7 bf a3 00 01 5c 31 bb c1 08 00 45 40
0010	00 4a 00 00 40 00 3b 11 e2 ea 4b 4b 4b 4b 18 06

Column	Value
Align Left	
Align Center	
Align Right	
Column Preferences...	
Edit Column	
Resize to Contents	
Resize Column to Width...	
Resolve Names	
No.	Number
Time	Time (format as specified)
Source	Source address
Destination	Destination address
Protocol	Protocol
Length	Packet length (bytes)
Time to live	ip.ttl
Coloring Rule Name	frame.coloring_rule.name
Host	http.host
TCP Delta	tcp.time_delta
Time since previous frame in this TCP stream	tcp.time_delta
HTTP time	text
Stream index	tcp.stream
Window size	tcp.window_size_value
Window size value	tcp.window_size_value
winsize	tcp.window_size
SN	tcp.seq
NextSN	tcp.nextseq
AckN	tcp.ack
SegLen	tcp.len
IP Flag	ip.flags.mf
Length	ip.len
Source Port	tcp.srcport
Time since previous frame in this TCP stream	tcp.time_delta
Fragment offset	ip.frag_offset
Info	Information
Remove this Column	

http-esp101.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000s	24.6.173.220	75.75.75.75	DNS	58	Standard query 0x0221 A www.espn.com
2	0.014749s	75.75.75.75	24.6.173.220	DNS	74	Standard query response 0x0221 A www.espn.com A 195
3	0.001121s	24.6.173.220	75.75.75.75	DNS	58	Standard query 0x8907 AAAA www.espn.com
4	0.013335s	75.75.75.75	24.6.173.220	DNS	108	Standard query response 0x8907 AAAA www.espn.com SC
5	0.001040s	24.6.173.220	199.181.132.250	TCP	52	19941 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4
6	0.031335s	199.181.132.250	24.6.173.220	TCP	52	80 → 19941 [SYN, ACK] Seq=0 Ack=1 Win=4380 Len=0 M

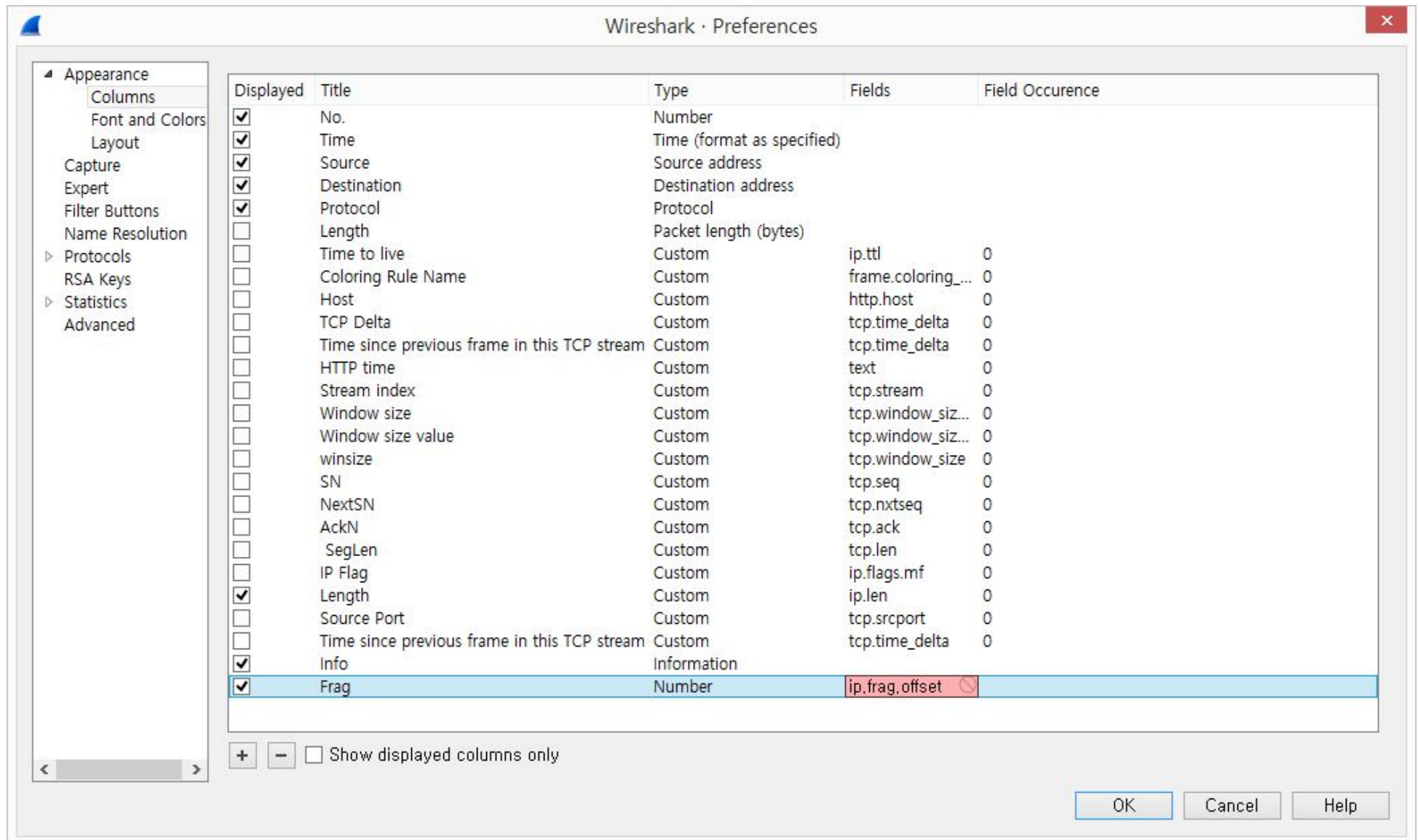
▶ Frame 1: 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface unknown, id 0
 ▶ Ethernet II, Src: HewlettP_a7:bf:a3 (d4:85:64:a7:bf:a3), Dst: Cadant_31:bb:c1 (00:01:5c:31:bb:c1)
 ▶ Internet Protocol Version 4, Src: 24.6.173.220, Dst: 75.75.75.75

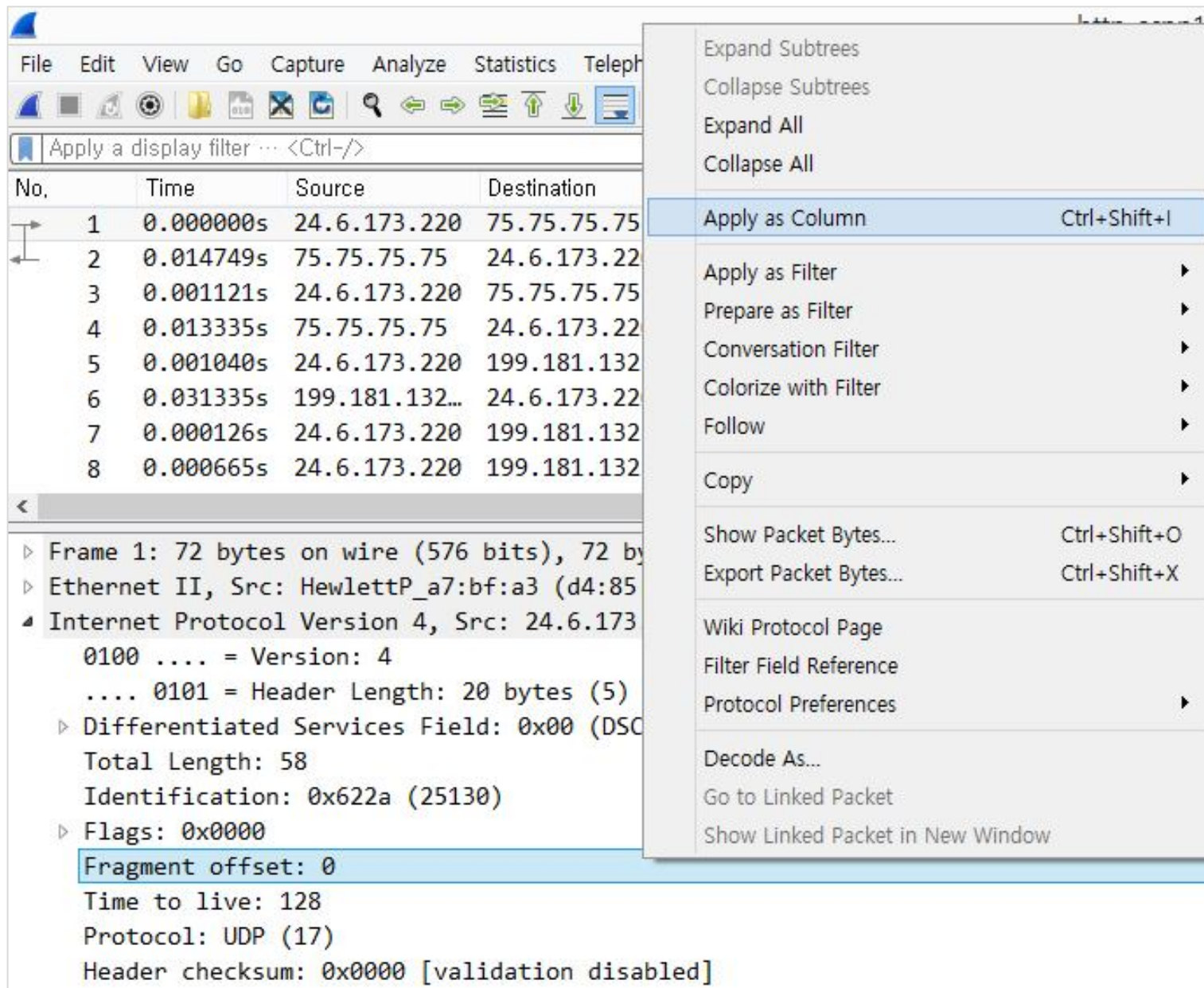
- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 58
- Identification: 0x622a (25130)
- ▶ Flags: 0x0000
- Fragment offset: 0
- Time to live: 128
- Protocol: UDP (17)
- Header checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
- Source: 24.6.173.220
- Destination: 75.75.75.75
- ▶ User Datagram Protocol, Src Port: 53184, Dst Port: 53
- ▶ Domain Name System (query)

Fragment offset (13 bits) (ip.frag_offset), 2 byte(s)
 Packets: 4900 · Displayed: 4900 (100.0%)
 Profile: Default

ip.frag_offset

Edit > Preference > Appearance > Columns

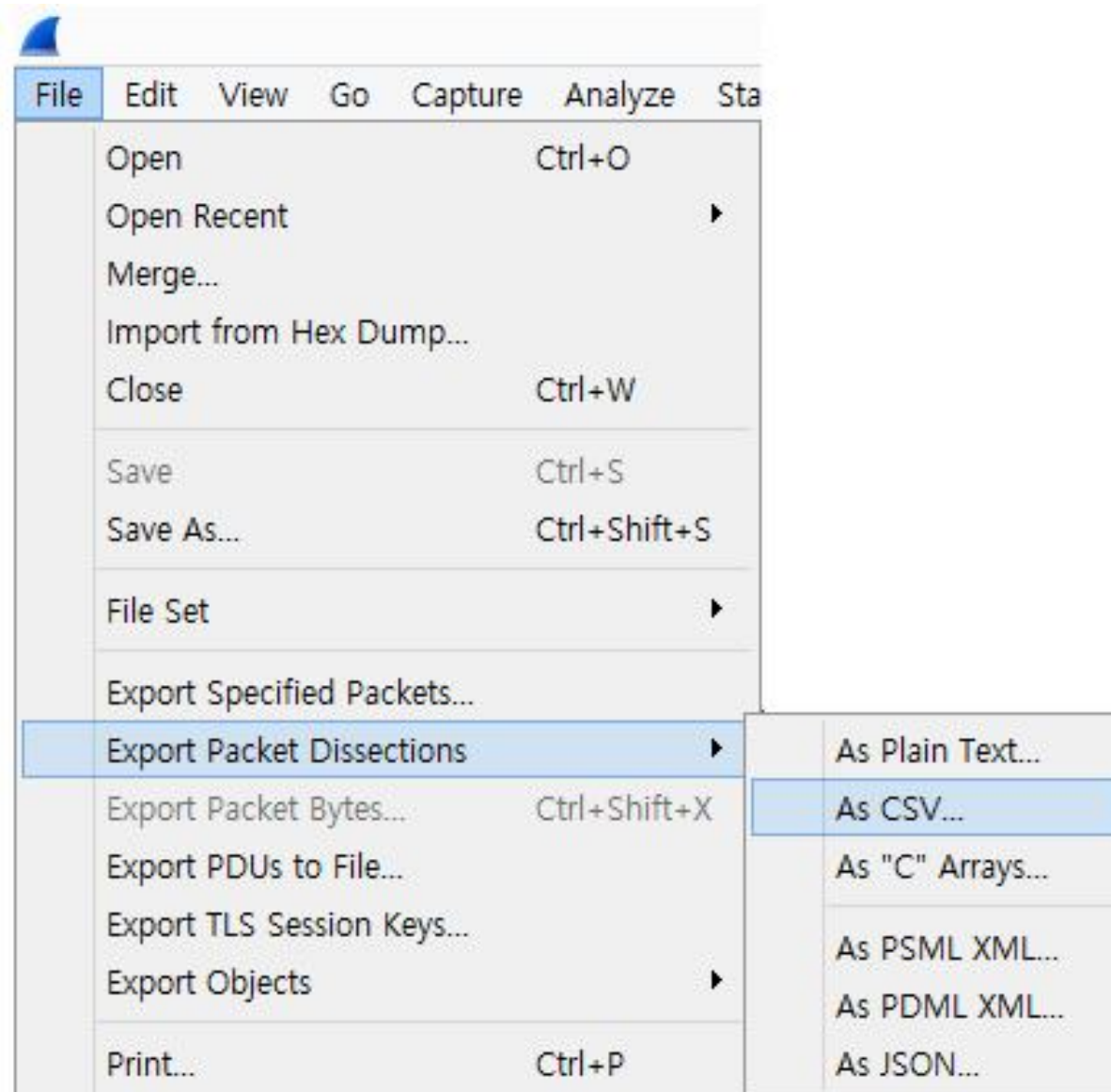


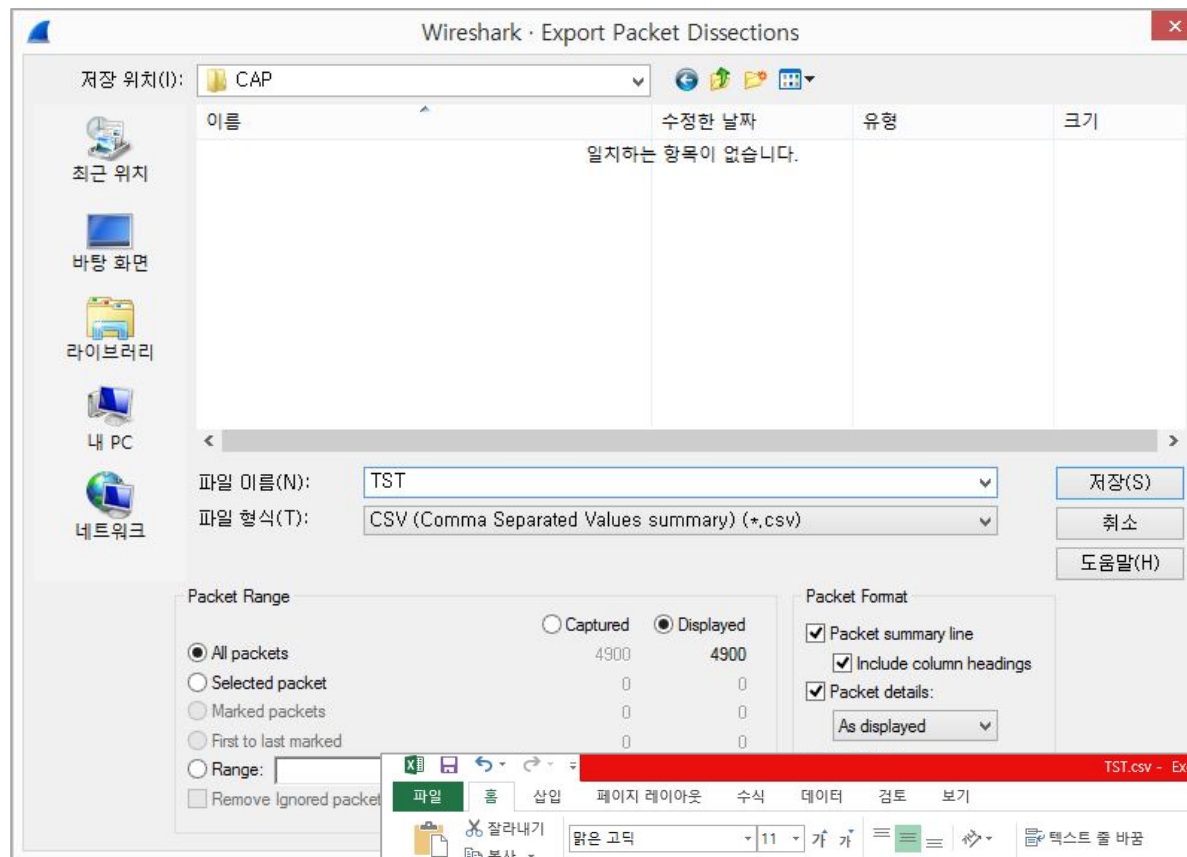


2) 컬럼 내용 정렬

No.	Time	Source	Destination	Protocol	Length	Info
3172	0.036831s	64.95.73.13	24.6.173.220	HTTP	261	HTTP/1.1 200 OK (text/html)
3005	0.020986s	64.95.73.13	24.6.173.220	HTTP	261	HTTP/1.1 200 OK (text/html)
2674	0.031928s	75.75.75.75	24.6.173.220	DNS	258	Standard query response 0x64
548	0.000007s	184.84.222.48	24.6.173.220	HTTP	258	Continuation
2616	0.000152s	75.75.75.75	24.6.173.220	DNS	246	Standard query response 0xd7
2614	0.012801s	75.75.75.75	24.6.173.220	DNS	246	Standard query response 0x33
4657	0.000002s	68.71.216.176	24.6.173.220	HTTP	245	Continuation
1389	0.000002s	184.84.222.120	24.6.173.220	HTTP	240	Continuation
662	0.000003s	184.84.222.152	24.6.173.220	HTTP	240	Continuation
759	0.033627s	74.125.224.59	24.6.173.220	HTTP	237	HTTP/1.1 302 Moved Temporari
373	0.000755s	70.42.13.100	24.6.173.220	HTTP	236	HTTP/1.1 200 OK
4733	0.826794s	68.71.216.157	24.6.173.220	HTTP	234	Continuation
2610	0.026348s	75.75.75.75	24.6.173.220	DNS	232	Standard query response 0x05
325	0.001912s	70.42.13.100	24.6.173.220	HTTP	231	HTTP/1.1 200 OK
4602	0.089654s	50.17.254.18	24.6.173.220	HTTP	223	HTTP/1.1 200 OK
4594	0.092823s	50.17.254.18	24.6.173.220	HTTP	223	HTTP/1.1 200 OK
2981	0.000001s	50.17.254.18	24.6.173.220	HTTP	223	HTTP/1.1 200 OK
2655	0.000490s	24.6.173.220	96.17.110.102	HTTP	217	POST /open/1 HTTP/1.1
2612	0.000386s	75.75.75.75	24.6.173.220	DNS	216	Standard query response 0x68
2707	0.000157s	75.75.75.75	24.6.173.220	DNS	213	Standard query response 0xa2
1100	0.000078s	75.75.75.75	24.6.173.220	DNS	211	Standard query response 0xa4

3) 컬럼 데이터 내보내기





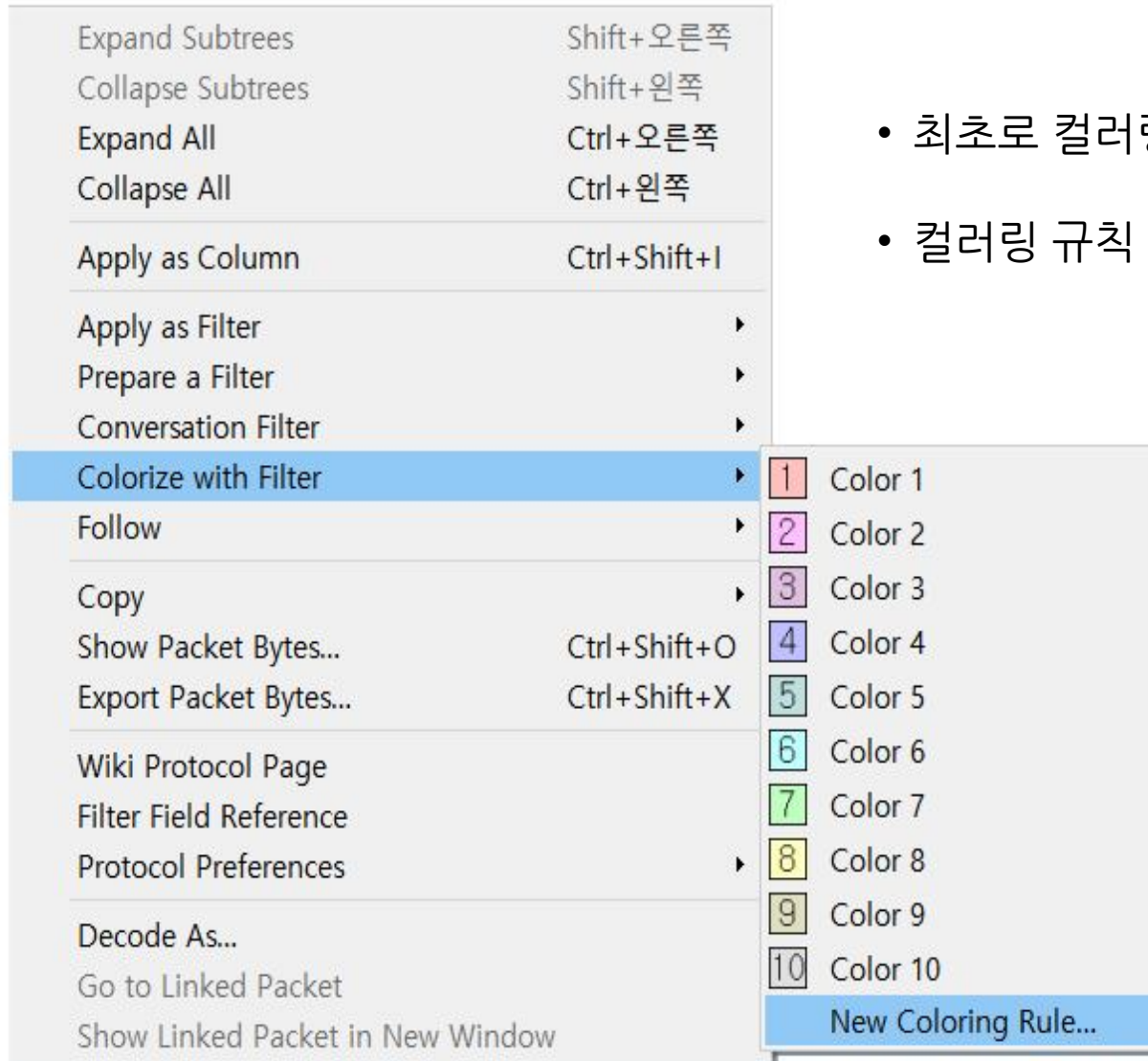
TST.csv - Excel

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000s	24.6.173.220	75.75.75.75	DNS	58	Standard query 0x0221 A www.espn.com
2	0.014749s	75.75.75.75	24.6.173.220	DNS	74	Standard query response 0x0221 A www.espn.com A 199.181.132.250
3	0.001121s	24.6.173.220	75.75.75.75	DNS	58	Standard query 0x8907 AAAA www.espn.com
4	0.013335s	75.75.75.75	24.6.173.220	DNS	108	Standard query response 0x8907 AAAA www.espn.com SOA dns-m1.espn.com
5	0.001040s	24.6.173.220	199.181.132.250	TCP	52	19941 > 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
6	0.031335s	199.181.132.250	24.6.173.220	TCP	52	80 > 19941 [SYN, ACK] Seq=0 Ack=1 Win=4380 Len=0 MSS=1460 WS=1 SACK_PERM=1
7	0.000126s	24.6.173.220	199.181.132.250	TCP	40	19941 > 80 [ACK] Seq=1 Ack=1 Win=65700 Len=0
8	0.000665s	24.6.173.220	199.181.132.250	HTTP	589	GET / HTTP/1.1
9	0.041099s	199.181.132.250	24.6.173.220	HTTP	470	HTTP/1.1 301 Moved Permanently (text/html)
10	0.002063s	24.6.173.220	75.75.75.75	DNS	57	Standard query 0xec2d A espn.go.com
11	0.012477s	75.75.75.75	24.6.173.220	DNS	73	Standard query response 0xec2d A espn.go.com A 68.71.216.176

2.4 검사합 오류 컬러링 규칙 해제

- 개별 컬러링 규칙 비활성화
- 모든 패킷 컬러링 비활성화

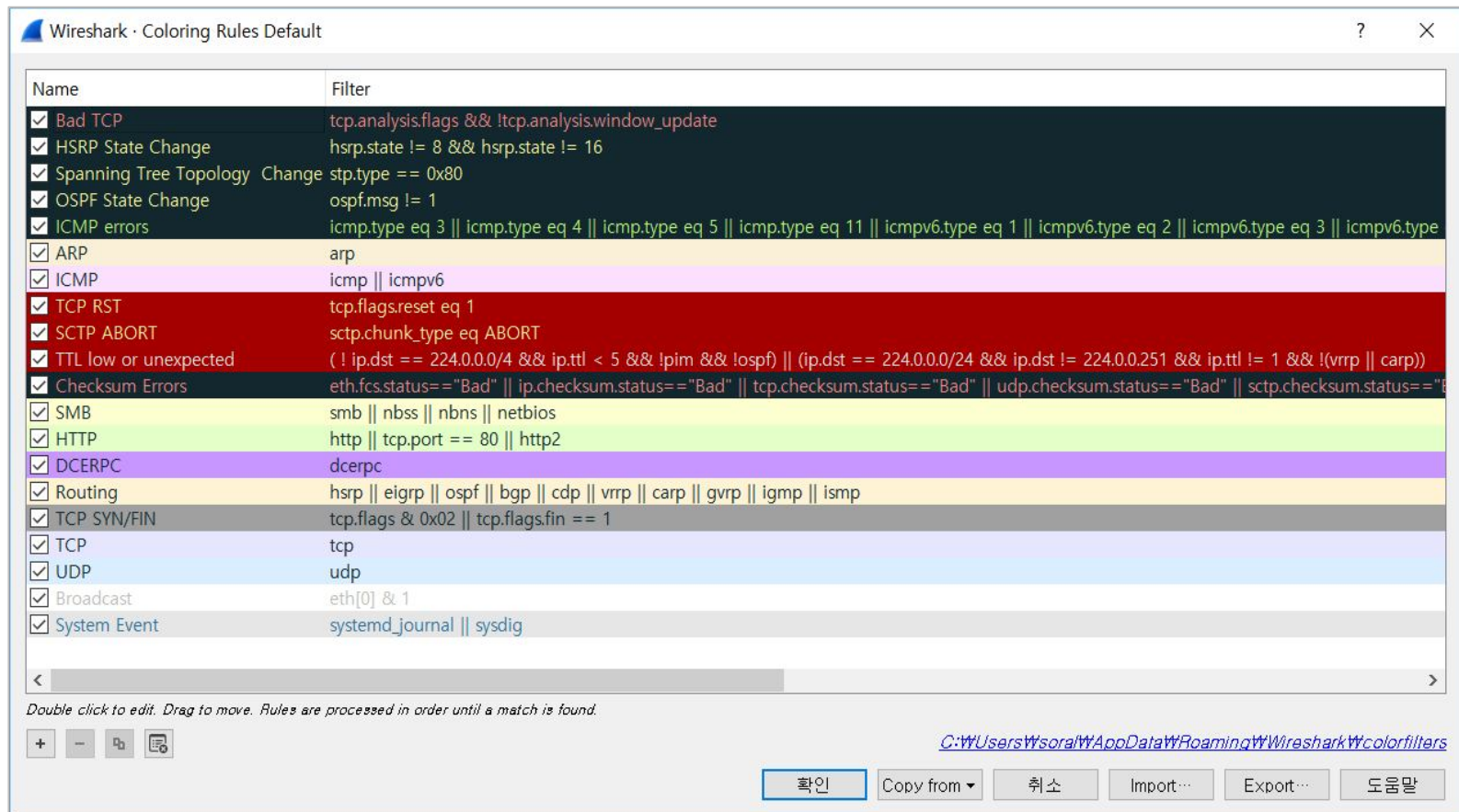
3) 컬러링 규칙 적용



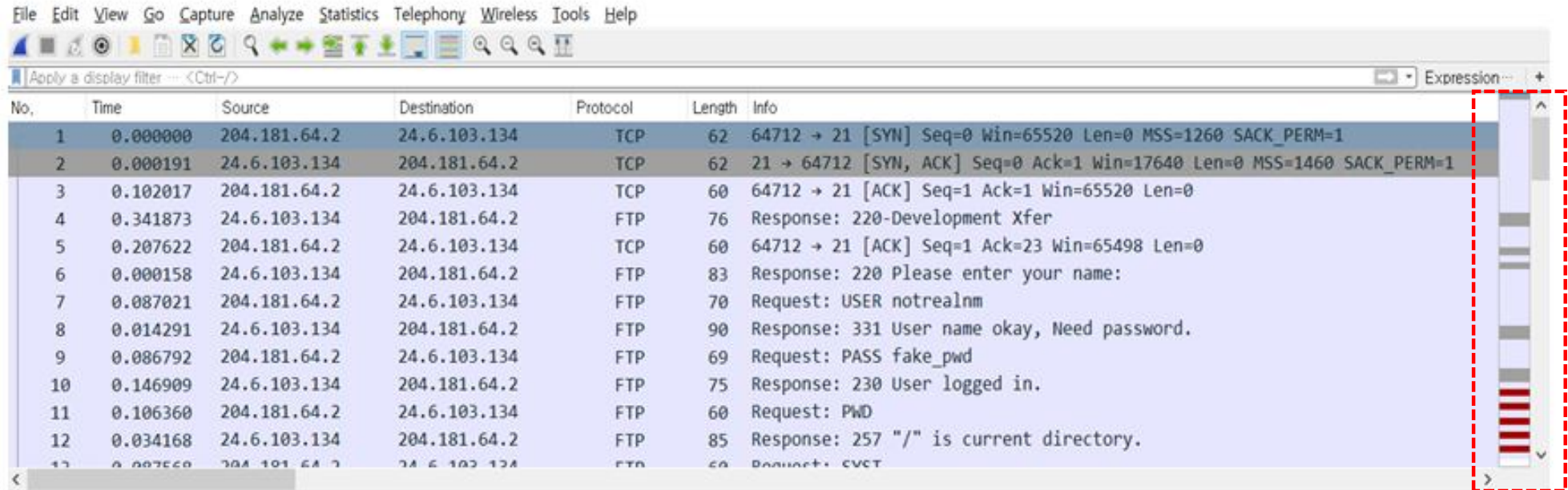
- 최초로 컬러링 규칙 생성
- 컬러링 규칙 생성에 오른쪽 클릭 방법 사용

4) View > Coloring Rules(색상규칙)

- 새로운 컬러링 규칙은 규칙 집합의 맨 위에 배치
- 컬러링 규칙의 배치는 중요
- 컬러링 규칙은 위에서 아래로 순서대로 처리
- 첫 번째 일치하는 컬러링 규칙에 따라 컬러링이 지정

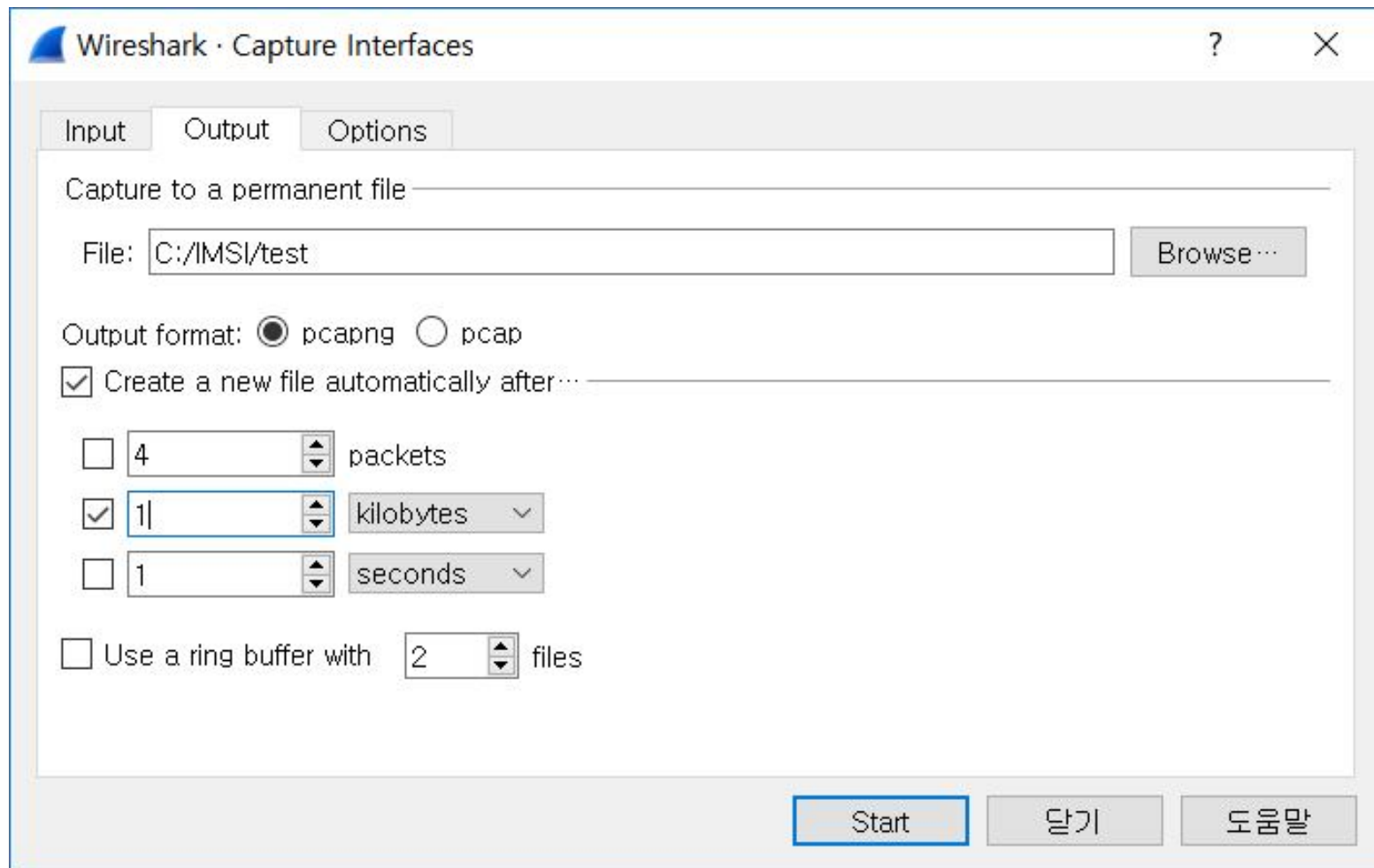


5) 지능형 스크롤바 마스터



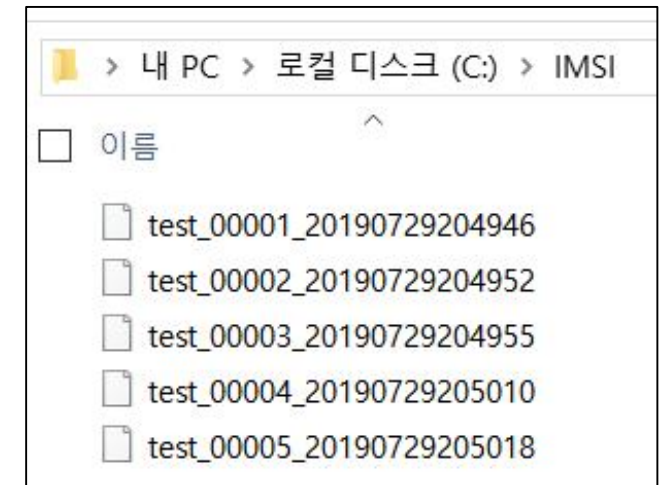
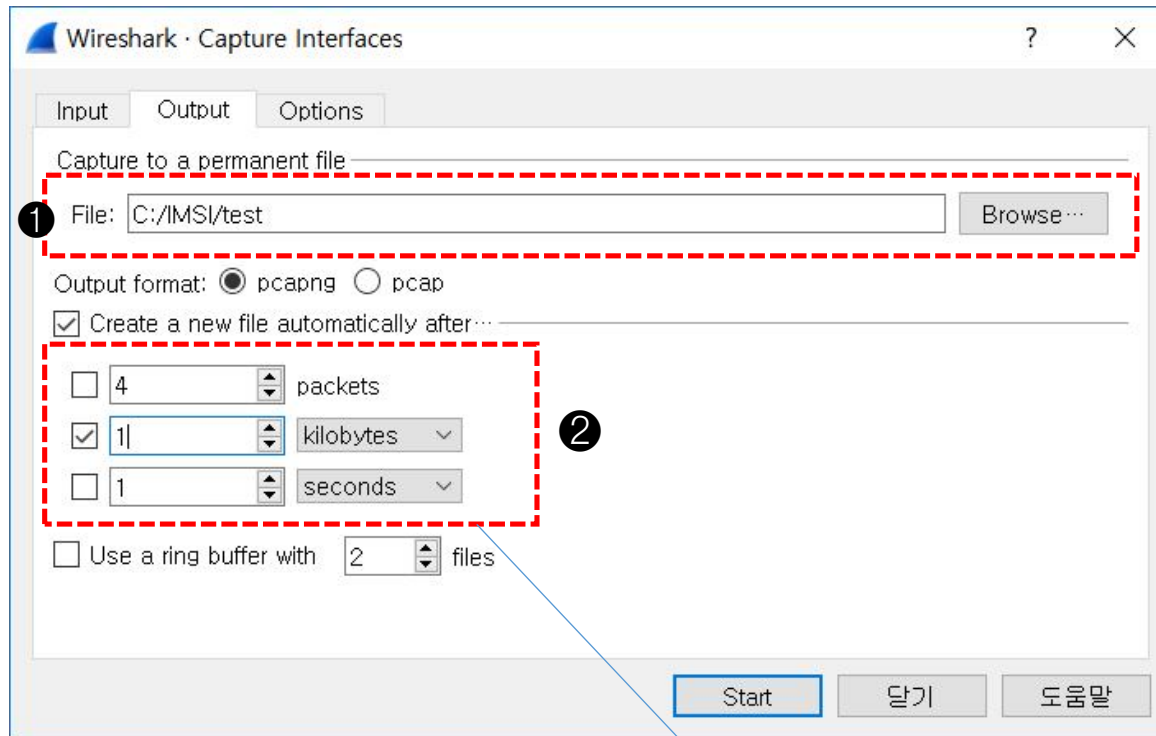
2.5 파일 집합으로 수집

- Capture Options > Output Tab > Create a new file automatically after...



파일 집합으로 수집

- Capture Options > Output Tab > Create a new file automatically after...



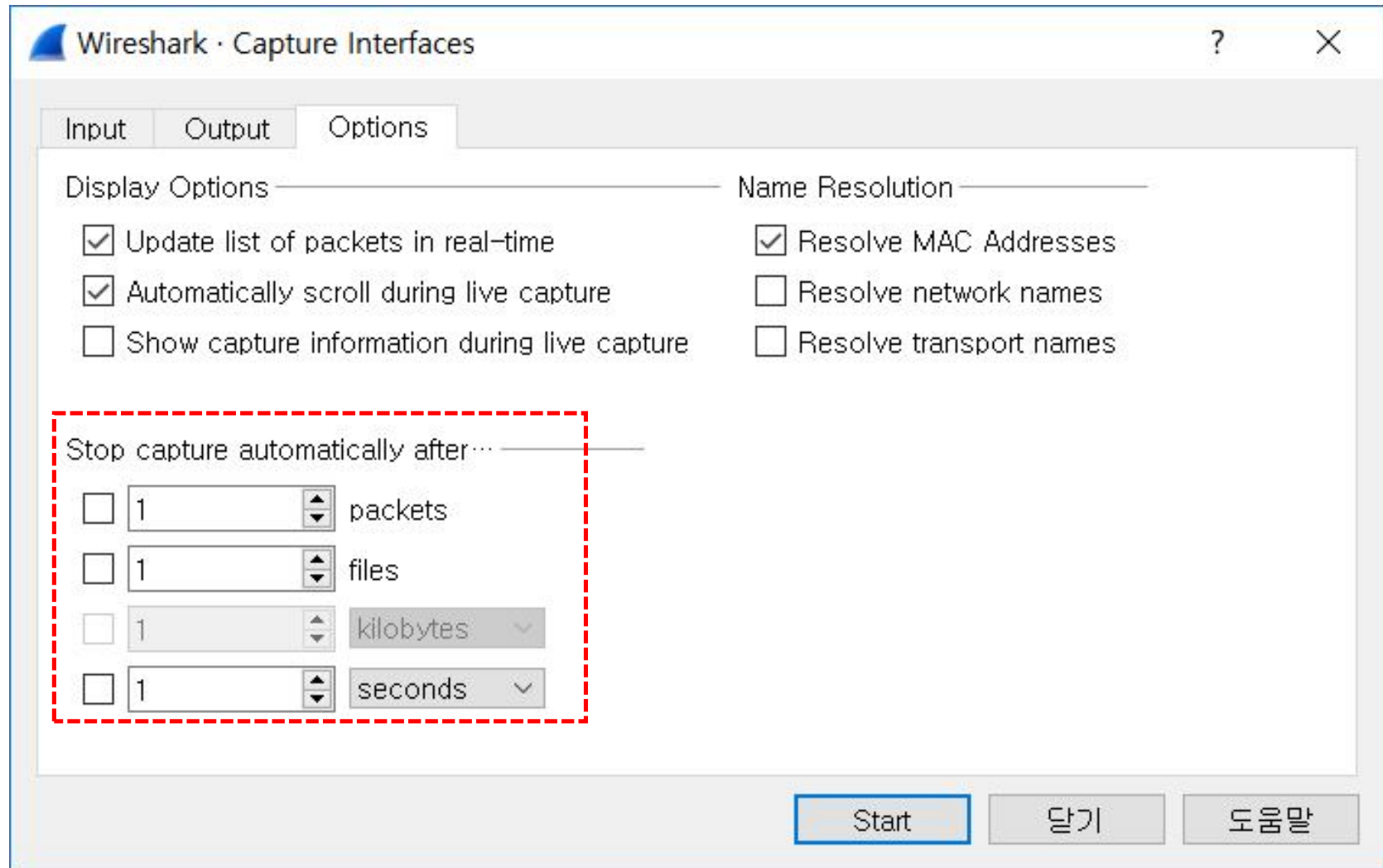
- File > Open
 - File > File set > List Files

파일당 4개의 패킷
1MB 파일 크기
1초마다

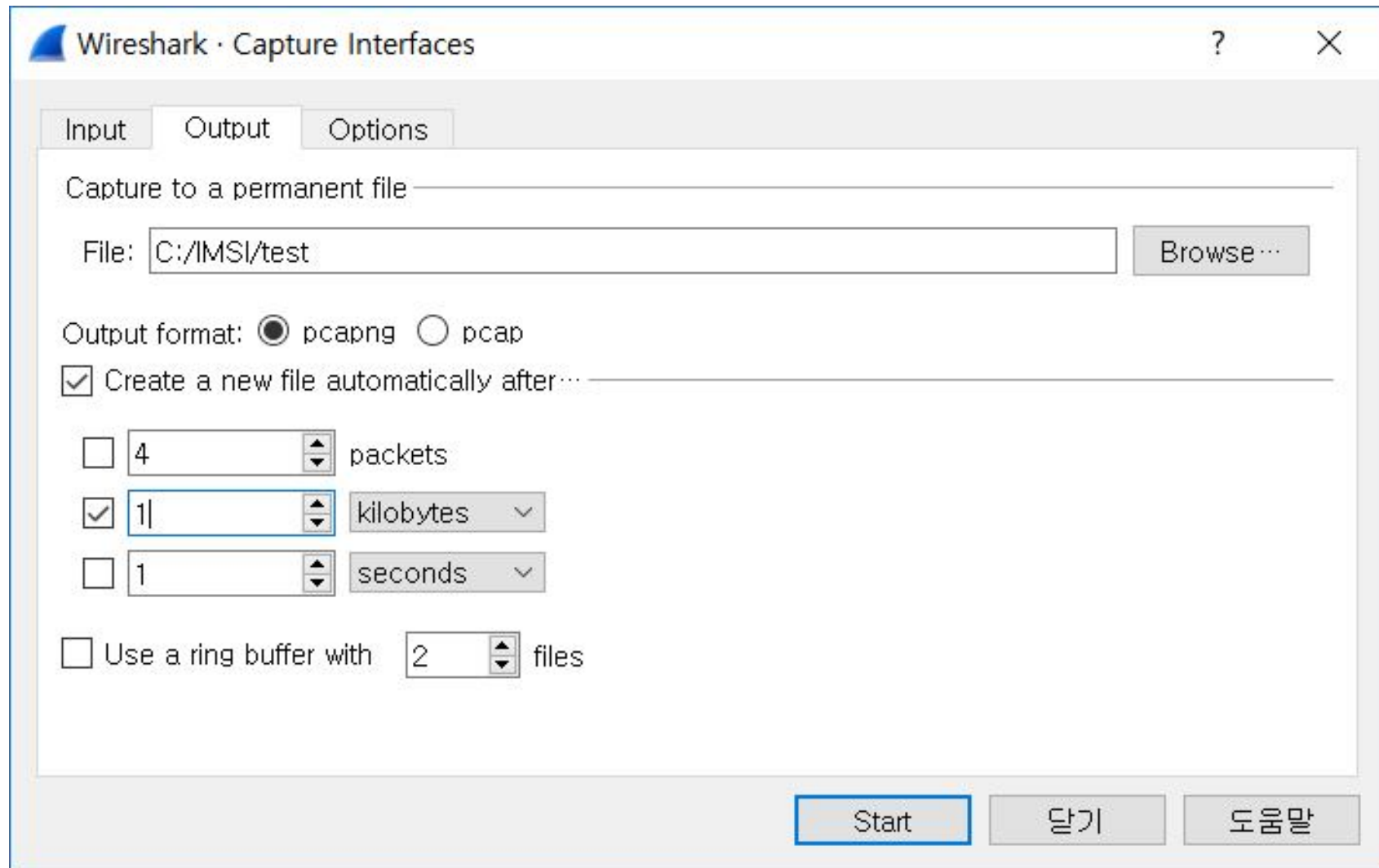
위의 어느 조건이든
먼저 만나면 파일 생성

파일 집합으로 수집

- Capture Options > Option Tab > Stop capture automatically after....



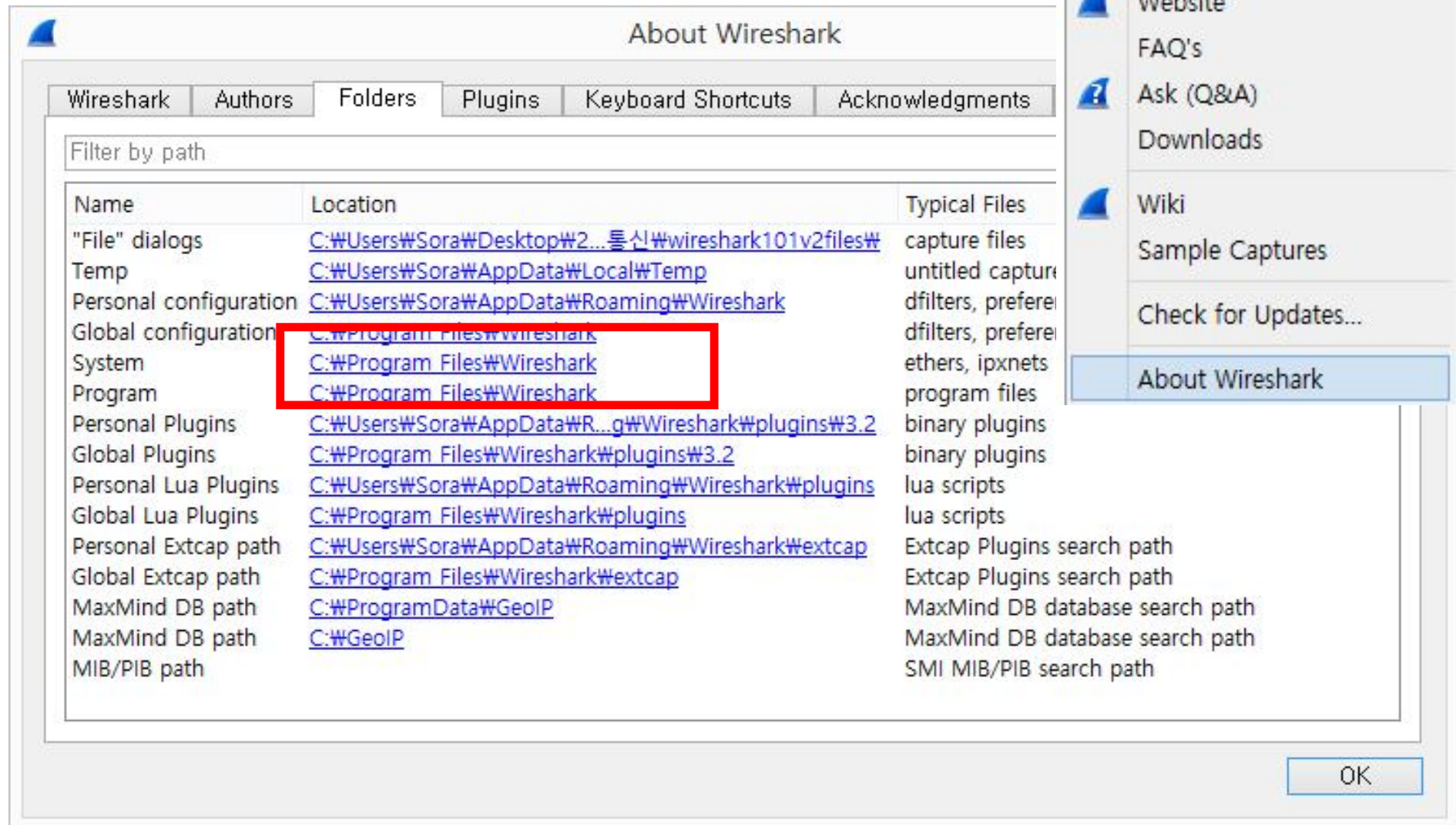
링 버퍼 사용

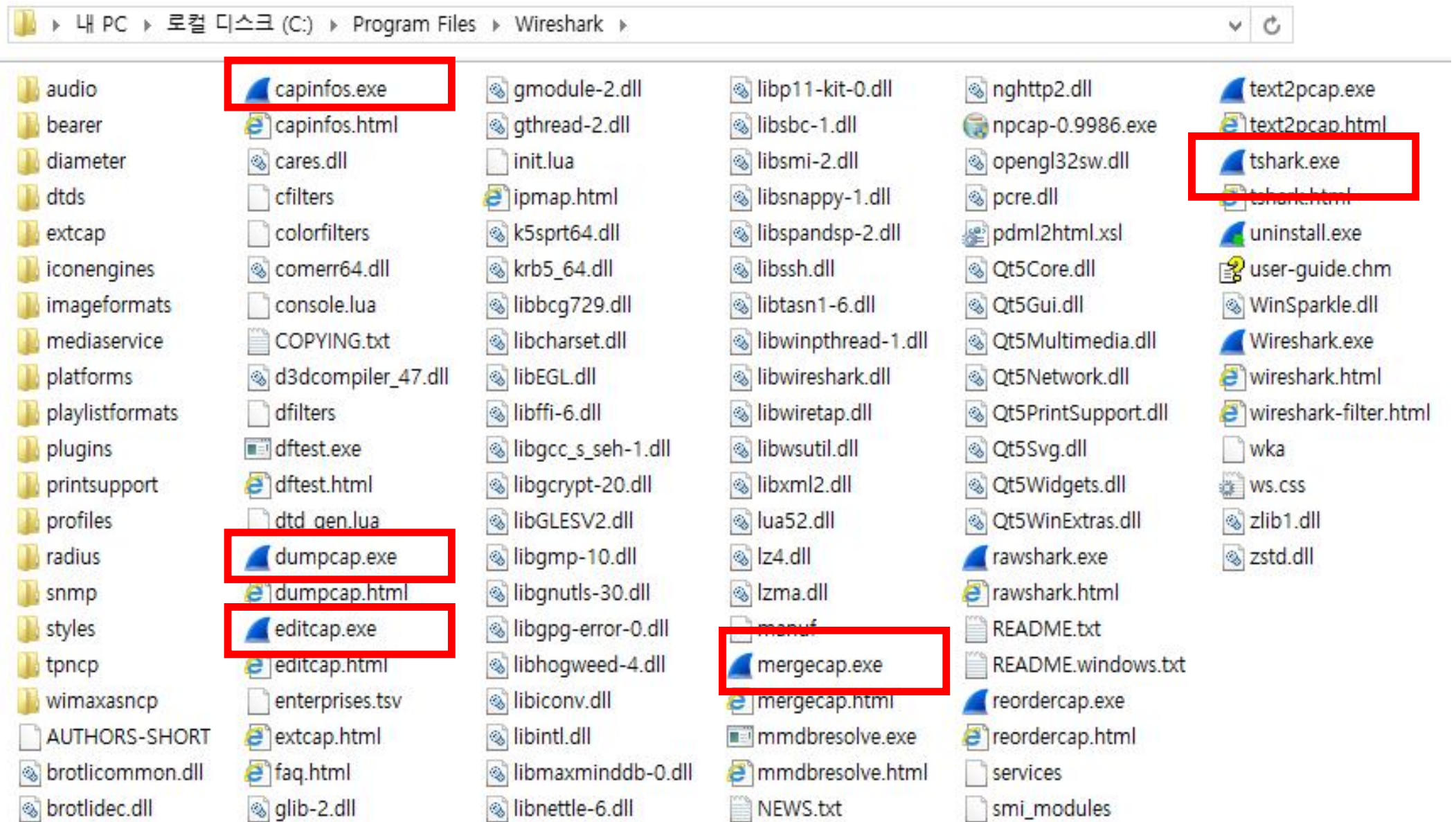


2.6 Tshark을 이용한 패킷 수집

	LINUX	MS-Window
패킷수집	TCPdump	TShark
패킷분석	Wireshark	Wireshark

Tshark을 이용한 패킷 수집





커맨드라인에서 트래픽 수집

- dumpcap.exe나 tshark.exe를 이용해 커맨드 라인으로 트래픽 수집
 - Tshark를 구동하면 dumpcap.exe를 호출해서 수집 기능을 활용

```
C:\Program Files\Wireshark>dir dumpcap.exe
```

```
C:\Program Files\Wireshark 디렉터리
```

```
2020-02-27 오전 05:47          420,416 dumpcap.exe
                  1개 파일          420,416 바이트
                  0개 디렉터리 86,740,180,992 바이트 남음
```

```
C:\Program Files\Wireshark>dir tshark.exe
```

```
C:\Program Files\Wireshark 디렉터리
```

```
2020-02-27 오전 05:47          582,720 tshark.exe
                  1개 파일          582,720 바이트
                  0개 디렉터리 86,740,180,992 바이트 남음
```

① tshark -h

```
C:\Program Files\Wireshark>tshark -h
TShark (Wireshark) 3.0.3 (v3.0.3-0-g6130b92b0ec6)
Dump and analyze network traffic.
See https://www.wireshark.org for more information.

Usage: tshark [options] ...
```

② tshark -D

```
C:\Program Files\Wireshark>tshark -D
1. \Device\NPF_{5256CF9B-1707-460C-B397-669CE852CFDE}
2. \Device\NPF_{A378EED4-AC87-4168-8B22-DFDD0B9EC62E}
3. \Device\NPF_{557124E1-28F6-4C2F-BDCE-1DFD75D011CE} (Wi-Fi)
4. \Device\NPF_{630C09BF-1CAB-457F-978E-7A0BBEE76CF5}
5. \Device\NPF_{E8E017EF-FC6D-4933-BD98-9AEC5CB26CF6}
6. \Device\NPF_{27056470-1B11-4C61-8E92-0B25B7CB57C9}
7. \Device\NPF_{DFBA2BCB-8989-40B4-873C-8CF862935F18}
8. \Device\NPF_{6EC32DD5-41B3-45FF-B701-BB15DC1E7E45}
9. \Device\NPF_{23E2556A-B636-483F-A13E-C3E889DD3825}
10. \Device\NPF_{415D90F5-DFA5-426C-ABFE-BCCAB1542370}
```

③ tshark -i 3

```
C:\Program Files\Wireshark>tshark -i 3
Capturing on 'Wi-Fi'
 1  0.000000 192.168.35.145 → 69.167.144.15 66 65160 64020 → https(443) [SYN] Seq=0 Win=65160 Len=0 MSS=1460 WS=256 SACK_PERM=1
 2  3.816065 192.168.35.145 → 121.53.202.218 66 65535 64021 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
 3  3.816070 192.168.35.145 → 121.53.202.218 66 65535 64022 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
 4  3.818776 121.53.202.218 → 192.168.35.145 66 8190 https(443) → 64022 [SYN, ACK] Seq=0 Ack=1 Win=8190 Len=0 MSS=1460 WS=16 SACK_PERM=1
 5  3.818895 192.168.35.145 → 121.53.202.218 54 1024 64022 → https(443) [ACK] Seq=1 Ack=1 Win=262144 Len=0
 6  3.819200 121.53.202.218 → 192.168.35.145 66 8190 https(443) → 64021 [SYN, ACK] Seq=0 Ack=1 Win=8190 Len=0 MSS=1460 WS=16 SACK_PERM=1
 7  3.819311 192.168.35.145 → 121.53.202.218 54 1024 64021 → https(443) [ACK] Seq=1 Ack=1 Win=262144 Len=0
 8  3.819713 192.168.35.145 → 121.53.202.218 295 1024 Client Hello
 9  3.819864 192.168.35.145 → 121.53.202.218 295 1024 Client Hello
```

* Ctrl+C 로 중단

④ tshark -i 3 -w TST.pcapng

```
C:\Program Files\Wireshark>tshark -i 3 -w TST.pcapng
Capturing on 'Wi-Fi'
2472

C:\Program Files\Wireshark>
```

⑤ tshark -r TST.pcapng

```
C:\Program Files\Wireshark>tshark -r TST.pcapng
 1  0.000000 192.168.35.145 → 69.167.144.15 74 65160 64159 → ht
 2  0.749769 192.168.35.145 → 104.74.232.184 54 256 64153 → ht
 3  0.752502 104.74.232.184 → 192.168.35.145 54 237 http(80) →
 4  0.752540 192.168.35.145 → 104.74.232.184 54 256 64153 → ht
 5  1.055385 192.168.35.145 → 110.76.141.124 66 64240 64160 → f
 6  1.058892 110.76.141.124 → 192.168.35.145 66 29200 https(443)
 7  1.058938 192.168.35.145 → 110.76.141.124 54 256 64160 → ht
 8  1.059047 192.168.35.145 → 110.76.141.124 256 256 Client Hel
```

⑥ tshark -i 3 -a files:3 -b duration:10 -w myshark.pcapng

```
C:\Program Files\Wireshark>tshark -i 3 -a files:3 -b duration:10 -w myshark.pcapng
Capturing on 'Wi-Fi'
4464
```

```
C:\Program Files\Wireshark>dir myshark*
C 드라이브의 볼륨에는 이름이 없습니다.
볼륨 일련 번호: BCF8-FA59
```

```
C:\Program Files\Wireshark 디렉터리
```

```
2018-08-12 오전 12:46          679,312 myshark_00001_20190801004608.pcapng
2018-08-12 오전 12:46      1,269,276 myshark_00002_20190801004618.pcapng
2018-08-12 오전 12:46      937,040 myshark_00003_20190801004629.pcapng
                3개 파일          2,885,628 바이트
                0개 디렉터리 126,970,032,128 바이트 남음
```

-a file:3 3개 파일 수집 후 자동 정지

-b duration:10 10초후에 다음 파일을 생성

-w myshark.pcapng 추적파일명

커맨드라인 수집 과정에서 수집 필터(캡처필터)

① `tshark -i 3 -f "tcp port 443" -w mysecpport443.pcapng`

```
C:\Program Files\Wireshark>tshark -i 3 -f "tcp port 443" -w mysecpport443.pcapng
Capturing on 'Wi-Fi'
2734
```

```
C:\Program Files\Wireshark>tshark -r mysecpport443.pcapng
 1  0.000000 192.168.35.145 → 69.167.144.15 74 65160 64805 → https(443) [SYN] Seq=0 W
 2  3.010265 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64805 → htt
 3  3.627403 192.168.35.145 → 52.175.23.79 66 64240 64806 → https(443) [SYN] Seq=0 Wi
 4  3.710985 52.175.23.79 → 192.168.35.145 66 8192 https(443) → 64806 [SYN, ACK] Seq=
 5  3.711076 192.168.35.145 → 52.175.23.79 54 258 64806 → https(443) [ACK] Seq=1 Ack=
 6  3.711563 192.168.35.145 → 52.175.23.79 274 258 Client Hello
 7  3.792328 52.175.23.79 → 192.168.35.145 1514 1026 [TCP segment of a reassembled PDU
 8  3.792330 52.175.23.79 → 192.168.35.145 1514 1026 https(443) → 64806 [ACK] Seq=146
 9  3.792335 52.175.23.79 → 192.168.35.145 538 1026 Server Hello, Certificate, Server
```

② `tshark -i 3 -f "tcp port 443 and host 192.168.1.1" -w my443.pcapng`

커맨드라인 수집 과정에서 디스플레이 필터

❶ **tshark -r “mysecport443.pcapng” -Y “tcp.analysis.flags”**

```
C:\Program Files\Wireshark>tshark -r "mysecport443.pcapng" -Y "tcp.analysis.flags"
  2    3.010265 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64805 → https(443)
1113   9.019700 192.168.35.145 → 69.167.144.15 66 65160 [TCP Retransmission] 64805 → https(443)
2734  24.038230 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64898 → https(443)
```

❷ **tshark -r “mysecport443.pcapng” -Y “tcp.analysis.flags” -w tcpflag.pcapng**

❸ **tshark -r “mysecport443.pcapng” -Y “http.request.method == GET”**